Reduced matrix sequences and Gaussian cubic estimates for the real Grothendieck constant

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Abstract

We construct an explicit sequence of rational matrices whose Grothendieck ratios converge to the reciprocal of a dimension-uniform Gaussian operator norm. Symmetry, zero diagonal, zero row and column sums, and doubly stochastic entrywise absolute values are imposed simultaneously by exact ratio-preserving maps. An exhaustive reduced sequence has monotone ratios converging to the full constant. We prove a coupled fourth- and sixth-cumulant inequality for real cubic Gaussian chaos and complete scalar Stein certificates giving $\mathbb{E}|Q| < 0.847$ whenever $\mathbb{E}Q^2 = 1$. Independently replayed continuous physical-source certificates then give

$$\sup_d \|P_1 - (711/1000)P_3\|_{L^\infty(\gamma_d)\to L^1(\gamma_d)} \le \frac{112782853}{200000000}.$$

The explicit reduced matrix sequence consequently proves $G_{\mathbb{R}} \ge 200000000/112782853 = 1.7733192119\ldots$, with quantified finite-matrix errors. A new global Hermite contact majorant retains the longitudinal cubic moment for further refinement. At $b = 1/2$, a published sharp operator inequality gives the unconditional benchmark $6\pi/11$. The target 1.775 and a three-significant-digit determination remain open.

1 Introduction and the numerical target

For a nonzero finite real matrix A, define the bilinear norms

$$C(A) = \max_{s_i,t_j\in\{-1,1\}} \sum_{ij} A_{ij}s_it_j, \qquad V(A) = \max_{\|u_i\|=\|v_j\|=1} \sum_{ij} A_{ij}\langle u_i, v_j\rangle.$$

The vector dimension is arbitrary; finite dimension $m + n$ suffices for an m-by-n matrix. Write $G(A) = V(A)/C(A)$ and $G_{\mathbb{R}} = \sup_A G(A)$.

Saha et al. [1] prove the unconditional bounds

$$\frac{6\pi}{11} \le G_{\mathbb{R}} \le \frac{\pi}{2\log(1+\sqrt{2})} - 3.47\cdot 10^{-4}.$$

Corollary 2 gives an exactly evaluable sequence attaining the published lower endpoint. The fresh complete norm proof below also gives the unconditional constructive bound

$$G_{\mathbb{R}} \ge \frac{200000000}{112782853} = 1.7733192119\ldots.$$

The earlier internal September 30 project estimate is retained solely as the all-dimension premise

$$\mathcal{N}(699/1000) := \sup_{d\ge 1} \|P_1 - (699/1000)P_3\|_{L_\infty(\gamma_d)\to L_1(\gamma_d)} \le \frac{1000}{1773}. \tag{1}$$

The earlier report attributes this estimate to a 55-interval amplitude cover and inherited integrations [2]. That complete cover has not been reconstructed or independently replayed here. Assuming (1) gives $G_{\mathbb{R}} \geq 1.773$; it is not an unconditional conclusion of this continuation.

The aim is a complete estimate $\mathcal{N}(b) \leq 40/71$ for a single coefficient b, since its reciprocal is 1.775. Together with the published upper bound above, that would determine $G_{\mathbb{R}}$ to three significant digits as 1.78. The published enclosure and the putative stronger project enclosure both cross the rounding boundary. A three-decimal lower bound is not a three-decimal approximation of the constant.

The principal results proved in this paper are the global constructive lower bound, the exact reduced sequence in Section 3, the coupled cumulant theorem in Section 5, the uniform cubic absolute-moment theorem in Section 6, and the support–source comparisons in Section 7. We also prove a sharper conditional Gaussian profile and a mixed cubic penalty in Section 8, and certify continuous operator regions in Section 9. Section 10 retains the actual cubic projection in a residual comparison and gives an exact transport between source amplitudes. Their proofs and numerical inputs do not require the missing historical amplitude data underlying (1). The use of that earlier operator norm estimate is confined to the stated corollary for $b = .699$.

2 The exact PAQ reduction

The right transformation is denoted R here to distinguish it from a Gaussian cubic Q. Suppose distinct columns of P have disjoint supports, and distinct rows of R have disjoint supports. Put

$$p_i = \sum_a |P_{ai}|, \qquad r_j = \sum_b |R_{jb}|.$$

Then

$$C(PAR) = C(D_p A D_r), \qquad V(PAR) = V(D_p A D_r). \tag{2}$$

In particular, if $p_i = \alpha > 0$ for every original row and $r_j = \beta > 0$ for every original column, both norms are multiplied by $\alpha\beta$, and $G(PAR) = G(A)$.

For scalar signs, the aggregate in row group i lies in $[-p_i, p_i]$. The maximum of the resulting bilinear form on the product of these intervals occurs at a vertex, and every vertex is attainable by aligned copied signs. For vectors the aggregate lies in the ball of radius p_i; aligned copies attain its boundary. Ball vectors may be completed to unit vectors using separate left and right orthogonal coordinates without changing cross inner products. This proves (2) for both norms.

Signed permutations and common scaling are special cases. Ordinary diagonal balancing and arbitrary orthogonal changes of coordinates are not consequences of this theorem. The requirements concern total absolute copying weights, not the number of copies.

2.1 A universal absolute doubly stochastic class

Let Z be a nonzero integer matrix with zero rows and columns removed. Set

$$r_i = \sum_j |Z_{ij}|, \qquad c_j = \sum_i |Z_{ij}|, \qquad L = \sum_i r_i = \sum_j c_j.$$

Make r_i copies of row i, and c_j copies of column j, and define the square matrix of order L by

$$\boxed{B_{(i,a),(j,b)} = \frac{Z_{ij}}{r_i c_j}, \quad 1 \leq a \leq r_i, \quad 1 \leq b \leq c_j.} \tag{3}$$

Each original row or column has total copying weight one. Thus

$$C(B) = C(Z), \qquad V(B) = V(Z), \qquad G(B) = G(Z). \tag{4}$$

Every row and column of $|B|$ sums to one. For example, a cloned row has sum $\sum_j c_j|Z_{ij}|/(r_i c_j) = 1$.

For rational input first clear its denominator. For real input choose rational approximations; both norms satisfy

$$|C(A) - C(A')|,\ |V(A) - V(A')| \le \sum_{ij} |A_{ij} - A'_{ij}|. \tag{5}$$

Consequently the supremum defining $G_{\mathbb{R}}$ is unchanged by restricting to rational matrices whose entrywise absolute values are doubly stochastic, over all orders. The clone order can be enormous, but both objectives can be evaluated on the smaller seed Z.

2.2 Symmetry, zero diagonal and zero sums

For B in (3), set

$$K(B) = \begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix}, \qquad J_0 = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad H(B) = J_0 \otimes K(B). \tag{6}$$

Then $H(B)$ is rational, symmetric, zero-diagonal, has zero row and column sums, and $|H(B)|$ is doubly stochastic. Its order is $4L$ and

$$C(H(B)) = 4C(B), \qquad V(H(B)) = 4V(B). \tag{7}$$

The two off-diagonal terms in $K(B)$ use independent left/right variable lists and can optimize simultaneously, giving factor two. The signed two-copy map J_0 gives a further factor two by (2). Hence all these restrictions can be imposed simultaneously without changing the supremum or any individual rational seed ratio.

2.3 A monotone sequence converging to the full constant

The analytic cubic sequence below has limit $1/\mathcal{N}(b)$. The reduction also permits an unconditional sequence converging to the full Grothendieck constant, without assuming that a cubic operator is optimal.

Proposition 1. *Let $\mathcal{A}_n$ consist of all nonzero real matrices of dimensions at most n whose entries lie in $2^{-n}\mathbb{Z} \cap [-1, 1]$. Choose a matrix $S_n \in \mathcal{A}_n$ maximizing G, with a fixed lexicographic rule for ties. Clear its dyadic denominator and apply* (3) *and* (6)*, obtaining $\widetilde{H}_n$. Then every $\widetilde{H}_n$ belongs to the simultaneous reduced class and*

$$G(\widetilde{H}_n) \le G(\widetilde{H}_{n+1}), \qquad \lim_{n\to\infty} G(\widetilde{H}_n) = G_{\mathbb{R}}.$$

Proof. Each candidate set is finite and the sets are nested. Both norms are positive on a nonzero matrix, and the exact reductions preserve their ratio, so the maxima exist and increase monotonically. For any fixed nonzero real matrix A, homogeneity permits the normalization $\max_{ij} |A_{ij}| \le 1$. Its dyadic approximants $A^{(n)}$ belong to $\mathcal{A}_n$ for sufficiently large n and have total entry error tending to zero. The two norms satisfy

$$|C(A) - C(B)| \le \sum_{ij} |A_{ij} - B_{ij}|, \qquad |V(A) - V(B)| \le \sum_{ij} |A_{ij} - B_{ij}|.$$

Consequently $G(A^{(n)}) \to G(A)$ and $\liminf_n G(\widetilde{H}_n) \ge G(A)$. Taking the supremum over A proves the asserted limit; the reverse inequality follows from the definition of $G_{\mathbb{R}}$. □

This is a finite exact specification at every stage. The scalar norm is computed by sign enumeration. The vector norm is the maximum of the rational linear functional $\sum_{ij} A_{ij}X_{i,m+j}$ on the compact set of positive semidefinite $(m+n)$-by-$(m+n)$ matrices with diagonal one. Its polynomial description makes exact algebraic optimization and comparison possible. Enumeration is computationally expensive. The analytic subfamily gives controlled candidates and a quantitative limit, while retaining a best previous candidate prevents a weaker trial from replacing it.

3 An explicit analytic seed and its limit

Let $b \in [0,1)$ be computable (a rational coefficient suffices), and put $T_b = P_1 - bP_3$, where P_j projects onto total Hermite degree j in Gaussian space. Start at $n \geq 2$ and let $d_n = n^2$. In each coordinate partition the line by the grid of mesh n^{-2} from $-n$ to n, together with its two exterior intervals. The grid has zero as an edge and is antipodally symmetric. Form product cells and retain the cells whose first coordinate lies on the negative half-line. These represent all antipodal pairs.

For a normalized product Hermite H_ξ, define

$$F_{j,n}[C,\xi] = 2\int_C H_\xi \, d\gamma_{d_n}, \qquad |\xi| = j, \quad j \in \{1,3\}.$$

The finite seed is

$$A_n(b) = F_{1,n}F_{1,n}^T - bF_{3,n}F_{3,n}^T. \tag{8}$$

All entries are analytic and computable. With normalized one-variable Hermites $h_k = \mathrm{He}_k/\sqrt{k!}$,

$$\int_a^b h_0 \, d\gamma = \Phi(b) - \Phi(a), \qquad \int_a^b h_k \, d\gamma = \frac{\varphi(a)\mathrm{He}_{k-1}(a) - \varphi(b)\mathrm{He}_{k-1}(b)}{\sqrt{k!}} \quad (k \geq 1),$$

with boundary terms zero at infinite endpoints. Products give the multivariate entries, so there is no unevaluated high-dimensional integral.

Extending cell signs oddly shows $C(A_n(b)) \leq \mathcal{N}(b)$, where $\mathcal{N}(b)$ is the supremum of the scalar operator norm over all Gaussian dimensions. Conversely any bounded odd functions in a fixed dimension can be approximated by these partitions in their first coordinates as $n \to \infty$. Conditional cell averaging converges in L_2; the remaining coordinates integrate out. The partitions need not be nested: continuous compactly supported functions are dense in L_2, and their cell-average error tends to zero as the mesh shrinks and the cutoff grows. Conditional averaging is an L_2 contraction, so density gives the same conclusion for arbitrary L_2 functions. Therefore

$$C(A_n(b)) \longrightarrow \mathcal{N}(b). \tag{9}$$

The supremum can be restricted to odd functions because T_b annihilates the even part. Maximization over bounded step functions reduces to signs at the vertices of the finite cubes.

For $X \sim \gamma_d$, the radial vector field $X/\|X\|$ gives

$$q_d(b) = a_d\left(1 - \frac{b}{2(d+2)}\right), \qquad a_d = \frac{2\Gamma((d+1)/2)^2}{d\Gamma(d/2)^2} \longrightarrow 1. \tag{10}$$

Its total degree-one energy is a_d; its degree-three energy is $a_d/[2(d+2)]$, by Gaussian radial integration. The ordinary L_2 norm of T_b is one. Consequently $V(A_n(b)) \leq 1$.

For an explicit approximation bound, let $r_n(X)$ be the vector of cell midpoints, with exterior coordinates clipped to $\pm n$, and set $v_n = r_n/\|r_n\|$. All its coordinates are nonzero because zero is

an edge of the grid. Thus this is an odd unit vector field, constant on cells. With $d = n^2$, $h = n^{-2}$, and

$$p_n = \min\{1, 2n\varphi(n)\}, \qquad E_n = \sqrt{\frac{1}{n^2(n^2-2)} + 8n\varphi(n)},$$

the elementary normalization inequality and Gaussian Mills bound give

$$\|v_n - X/\|X\|\|_{L_2}^2 \le dh^2 \mathbb{E}\|X\|^{-2} + 4\Pr(\max_i |X_i| > n) \le E_n^2. \tag{11}$$

Here $\mathbb{E}\|X\|^{-2} = 1/(d-2)$, and the union bound gives $\Pr(\max_i |X_i| > n) \le 2d\varphi(n)/n$. In particular $E_n = n^{-2} + O(n^{-4})$.

The vector objective admits a sharper, quadratic error estimate. Write $R = \|X\|$, $\mu = \mathbb{E}R$, $w = X/R$, and $e = v_n - w$. The degree-one and degree-three projections above give

$$T_b w = t(R)w, \qquad t(R) = \frac{\mu R}{d}\left(1 - \frac{b}{2} + \frac{bR^2}{2(d+2)}\right) \ge 0.$$

Pointwise unit length implies $e \cdot w = -\|e\|^2/2$. Since the lowest odd spectral coefficient of T_b is $-b$,

$$\langle w, T_b w\rangle - \langle v_n, T_b v_n\rangle = \mathbb{E}[t(R)\|e\|^2] - \langle e, T_b e\rangle \le \mathbb{E}[(t(R) + b)\|e\|^2].$$

Off the exterior event, $\|e\|^2 \le dh^2/R^2$; on it, $\|e\|^2 \le 4$. Using $\mu\mathbb{E}R^{-1} = da_d/(d-1)$, $a_d \le 1$, and

$$\mathbb{E}[R\mathbf{1}_E] \le \sqrt{d\Pr(E)}, \qquad \mathbb{E}[R^3\mathbf{1}_E] \le \sqrt{d(d+2)(d+4)\Pr(E)},$$

proves the entirely explicit bound

$$\begin{aligned} L_n(b) = \frac{1}{n^2}&\left[\frac{1 - b/2}{n^2 - 1} + \frac{b}{2(n^2+2)} + \frac{b}{n^2-2}\right] \\ &+ 4\left[1 - \frac{b}{2} + \frac{b}{2}\sqrt{\frac{n^2+4}{n^2+2}}\right]\sqrt{p_n} + 4bp_n. \end{aligned} \tag{12}$$

Thus

$$V(A_n(b)) \ge q_{n^2}(b) - L_n(b) \longrightarrow 1, \qquad L_n(b) = (1+b)n^{-4} + O(n^{-6}). \tag{13}$$

The exponentially small terms in (12) are retained in every finite bound. The first-order direction error cancels because the radial field is pointwise aligned with its operator image.

3.1 The sequence inside the reduced class

Choose computable symmetric dyadic matrices $\widehat{A}_n(b)$ with total entry error $\eta_n \le n^{-4}$. Directed entry intervals select the first dyadic approximation whose certified error meets that budget; this avoids deciding an exact rounding tie. Clear the denominator to $Z_n(b)$, apply (3), and then (6):

$$\boxed{H_n(b) = H(B(Z_n(b)))}. \tag{14}$$

Every $H_n(b)$ lies in the simultaneous reduced class, and

$$G(H_n(b)) = G(\widehat{A}_n(b)) \ge \frac{q_{n^2}(b) - L_n(b) - \eta_n}{\mathcal{N}(b) + \eta_n}.$$

Indeed $C(A_n(b))$ is bounded away from zero: a first-coordinate hemisphere gives $(2/\pi)(1-b/6) > 5/(3\pi) > 1/4$. Thus $n \geq 2$ also ensures $\widehat{A}_n(b) \neq 0$. Equations (5)–(13) prove

$$\lim_{n\to\infty} G(H_n(b)) = \frac{1}{\mathcal{N}(b)}. \tag{15}$$

In particular (1) proves $\lim_n G(H_n(.699)) \geq 1.773$ assuming that premise. The exact normalization preserves this endpoint. The gamma recurrence gives $a_{d+2}/a_d = (d+1)^2/[d(d+2)]$, and consequently $a_d > 1 - 1/(2d)$: along each parity the quotient $a_d/(1-1/(2d))$ decreases to one. Therefore any proved $\mathcal{N}(b) \leq B$ gives the finite, computable lower bound

$$G(H_n(b)) \geq \frac{(1-1/(2n^2))(1-b/[2(n^2+2)]) - L_n(b) - n^{-4}}{B + n^{-4}} = \frac{1}{B} - \frac{1+b}{2Bn^2} + O(n^{-4}). \tag{16}$$

Corollary 2 (An exactly evaluable Gaussian benchmark)**.** *Let $T = P_1 - \frac{1}{2}P_3$ on standard Gaussian space and let A_n be its antipodally folded product-cell compression, with dimension $d = n^2$, mesh n^{-2}, cutoff $\pm n$, and 0 a coordinate-cell boundary. Then, for every $n \geq 2$,*

$$C(A_n) = \frac{11}{6\pi}, \qquad V(A_n) \leq 1, \qquad \lim_{n\to\infty} G(A_n) = \frac{6\pi}{11}.$$

Choose a symmetric rational approximation $\widehat{A}_n$ with total entrywise error at most $\eta_n = n^{-4}$, and apply the exact cloning, symmetric dilation and balancing reductions to obtain H_n. Thus H_n is symmetric, has zero diagonal and zero row sums, and $|H_n|$ is doubly stochastic. The same actual ratio limit holds:

$$\lim_{n\to\infty} G(H_n) = \frac{6\pi}{11} = 1.713595992867159948\ldots.$$

The rationalized ratios need not lie below their limiting value.

Proof. The all-odd-sign inequality proved by Saha et al. [1], Sections 8–9, is

$$2\langle P_1 f, P_1 g\rangle - \langle P_3 f, P_3 g\rangle \leq \frac{11}{6}\frac{2}{\pi}.$$

Replacing g by $-g$ gives the corresponding absolute bound. It also holds for odd functions of absolute value at most one: successively maximize the bilinear pairing in each function, choosing the odd sign of the other function's image under T. For $s(X) = \operatorname{sgn}(X_1)$ and $\nu = \sqrt{2/\pi}$, the raw Hermite polynomial $\mathrm{He}_3(z) = z^3 - 3z$ gives

$$P_1 s = \nu X_1, \qquad P_3 s = -\frac{\nu}{6}\mathrm{He}_3(X_1), \qquad \langle s, Ts\rangle = \nu^2 - \frac{\nu^2}{12} = \frac{11}{6\pi}.$$

The hemisphere sign is exactly constant on each product cell. Consequently its folded matrix assignment attains the external upper bound, establishing the exact scalar norm for every finite n. For any pair of unit-vector assignments, their odd cell fields v, w have $\int \|v\|^2\, d\gamma_d = \int \|w\|^2\, d\gamma_d = 1$; the L^2 operator norm $\|T\|_{2\to 2} = 1$ and Cauchy–Schwarz give $V(A_n) \leq 1$. The radial unit field and the established midpoint estimate give $V(A_n) \to 1$. Finally, total entrywise error bounds both norm errors by η_n, and every subsequent reduction preserves G exactly. This proves both asserted limits. □

For completeness, the following are explicit finite lower bounds. Put $d = n^2$, $p_n = 2n\varphi(n)$, and

$$e_n = \frac{1}{d}\left(\frac{3}{4(d-1)} + \frac{1}{4(d+2)} + \frac{1}{2(d-2)}\right)$$

$$+4\left(\frac{3}{4}+\frac{1}{4}\sqrt{\frac{d+4}{d+2}}\right)\sqrt{p_n}+2p_n,$$
$$q_n=\left(1-\frac{1}{2d}\right)\left(1-\frac{1}{4(d+2)}\right)-e_n.$$

The gamma-ratio lower bound $a_d > 1-1/(2d)$ and the quadratic midpoint loss imply

$$G(A_n)\geq\frac{q_n}{11/(6\pi)},\qquad G(H_n)\geq\frac{q_n-n^{-4}}{11/(6\pi)+n^{-4}}.$$

Directed MPFR256 evaluation, rounded down to twelve decimal places, gives

n	$G(A_n)$ lower bound	$G(H_n)$ lower bound
32	1.712339483749	1.712335051217
128	1.713517545081	1.713517527758
1024	1.713594767206	1.713594767202

These evaluations are recorded in `saha_benchmark_certificate.json`; the external scalar theorem is the sole new norm input and has not been independently replayed here. This benchmark does not invoke the retained, unverified global norm premise used for the earlier 1.773 candidate.

The same external theorem determines the complete analytic range $0 \leq b \leq 1/2$. Indeed $T_b = 2bT_{1/2} + (1-2b)P_1$, and $\mathcal{N}(0) = 2/\pi$. Convexity and the same hemisphere equality give

$$\mathcal{N}(b)=\frac{2}{\pi}(1-b/6),\qquad \lim_n G(H_n(b))=\frac{\pi}{2(1-b/6)}\quad(0\leq b\leq 1/2).$$

This exactly soluble range ends at $6\pi/11$; improving it requires $b > 1/2$ or a different operator family.

4 Coefficient choices and construction ceilings

The final certified sequence uses $b = 711/1000$, with the complete norm cap in Theorem 30. Reaching the stronger target requires a bound $\mathcal{N}(b) \leq 40/71$ at one fixed coefficient.

The initial candidate uses $b = 18/25 = .72$ in (8)–(14). For this coefficient, the target requires the all-dimension inequality

$$\boxed{\mathcal{N}(18/25)\leq\frac{40}{71}.}\qquad\frac{1}{40/71}=1.775. \tag{17}$$

The previous .699 operator cannot reach this target: its retained directed diffuse-cubic sign witness limits its reciprocal-norm construction to $1.77440101857\ldots$ [4]. Changing the coefficient is therefore necessary within the pure cubic family.

The .72 coefficient has a fresh directed lower witness. Let $h_3(x) = (x^3-3x)/\sqrt{6}$, let all Gaussians below be independent, and put

$$Y_N=N^{-1/2}\sum_{j=1}^{N}h_3(X_j),\qquad a=511/200,\quad c=393/1000,$$
$$f_N=\operatorname{sgn}(Y_N+aZ+cZ^3),\qquad g_N=\operatorname{sgn}(-Y_N+aZ+cZ^3).$$

The spectral central limit calculation assigns ordinary Z-degree j and diffuse-cubic degree k to total original Hermite degree $j+3k$. Its full degree accounting is proved in Appendix E. Writing $q(z)=az+cz^3$, the limiting moments are

$$m_Y = 2\int_{\mathbb{R}} \varphi(z)\varphi(q(z))\,dz, \quad m_1 = 2\int_{\mathbb{R}} \varphi(z)q'(z)\varphi(q(z))\,dz,$$

$$m_3 = \frac{2}{\sqrt{6}}\int_{\mathbb{R}} (z^2-1)\varphi(z)q'(z)\varphi(q(z))\,dz.$$

The scalar pairing is $m_1^2 - b(m_3^2 - m_Y^2)$. Directed Taylor integration and explicit tails give

$$\mathcal{N}(.72) > .563241882518393914882, \qquad \frac{1}{\mathcal{N}(.72)} < 1.7754361511. \tag{18}$$

This is a ceiling for the construction, not an achieved lower bound. The witness lies below $40/71$ by more than .00013839917, and hence does not rule out (17). Old whole-frame profile transfers fail to certify (17); their failure is a limitation of those sufficient estimates, not an obstruction from the true norm.

Proposition 3 (Exact reflected norm witness at $b = 7171/10000$)**.** *Let* $\mathcal{N}(b) = \sup_d \|P_1 - bP_3\|_{L^\infty(\gamma_d)\to L^1(\gamma_d)}$. *Then*

$$\mathcal{N}\left(\frac{7171}{10000}\right) > \frac{56323352479769}{10^{14}}.$$

Consequently the asymptotically unit-numerator reciprocal-norm construction for this fixed coefficient has ceiling strictly below $1775462496412/10^{12} = 1.775462496412$. *This is a construction ceiling, not a Grothendieck lower endpoint.*

Proof. Use the normalized Hermite $h_3(x) = (x^3-3x)/\sqrt{6}$. Set $a = 26412631/10^7$, $c = 4084047/10^7$, $q(z) = az + cz^3$ and $Y_n = n^{-1/2}\sum_{j=1}^n h_3(X_j)$ for independent standard Gaussian variables. Use the actual original-space signs $f_n = \operatorname{sgn}(Y_n + q(Z))$ and $g_n = \operatorname{sgn}(-Y_n + q(Z))$. The original-degree projection accounting for the reflected sign witness applies because $q'(z) \geq a > 0$; in particular the diffuse channel retains original degree three. Thus the limiting first- and third-chaos pairings are m_1^2 and $m_3^2 - m_Y^2$, respectively, where

$$m_Y = \frac{2}{\pi}\int_0^\infty e^{-(z^2+q(z)^2)/2}\,dz, \quad m_1 = \frac{2}{\pi}\int_0^\infty q'(z)e^{-(z^2+q(z)^2)/2}\,dz,$$

$$m_3 = \frac{2}{\pi\sqrt{6}}\int_0^\infty (z^2-1)q'(z)e^{-(z^2+q(z)^2)/2}\,dz.$$

The exact-rational directed certificate `certify_cubic7171_norm_witness.py` integrates Taylor terms through degree eight on 320 rational cells of $[0,5]$, bounds the degree-ten remainder over each entire cell, and bounds the omitted Gaussian tails explicitly. At 384-bit MPFR precision it encloses

$$m_1^2 - \frac{7171}{10000}(m_3^2 - m_Y^2) \in [0.563233524797690133779 3,\ 0.563233524797690134435 6].$$

Every finite-dimensional sign pairing is bounded above by $\mathcal{N}(7171/10000)$; taking the limit proves the assertion. The reciprocal enclosure has an upper endpoint below 1.77546249641158, proving the stated construction ceiling. The witness remains below $40/71$ and therefore does not exclude a future certified lower endpoint of 1.775. A separate Arb replay at 512 bits, Taylor order twelve and 400 rational cells recomputed all three integrals and Gaussian tails without importing the producer, independently accepting the stated inequalities. □

A fresh reflected-sign witness also excludes $b = 141/200 = .705$ from reaching the target within this reciprocal-norm family. Use the exact head $q(z) = (31091537/10^7)z + (49105809/10^8)z^3$ in the same construction. Directed integration gives

$$\mathcal{N}(.705) > \frac{563383848679}{10^{12}} > \frac{40}{71}, \qquad \frac{1}{\mathcal{N}(.705)} < 1.774988761828 < 1.775.$$

An independent 512-bit Arb replay, with Taylor order twelve and 400 rational cells, checks all three moments and the omitted Gaussian tails. The same replay at $b = .713$ gives a construction ceiling below 1.775407850864, leaving the target possible at the base coefficient $b = .713$. These are ceilings for the specified matrix limits, and impose no upper bound on $G_{\mathbb{R}}$.

5 A coupled cumulant inequality for real cubic chaos

Let $Q = I_3(f)$ be a real third Gaussian chaos, normalized by $\mathbb{E}Q^2 = 6\|f\|_F^2 = 1$, where f is a real fully symmetric third-order tensor. The flattening Gram matrices below are symmetric matrices; they need not be fully symmetric as four-index tensors. The norm $\|\cdot\|_F$ is the Euclidean norm of all entries. Define

$$\kappa = \mathbb{E}Q^4 - 3, \qquad \kappa_6 = \mathbb{E}Q^6 - 15\mathbb{E}Q^4 + 30, \qquad U = \tfrac{1}{3}\|DQ\|^2 - 1, \qquad K = 194/\sqrt{10}.$$

Theorem 4. *In every finite Gaussian dimension,*

$$\kappa_6 \le K\kappa^{3/2} - 633\sqrt{\kappa}\left(\mathbb{E}U^2 - \frac{7\kappa}{45}\right). \tag{19}$$

Also $0 < \kappa \le 90$ *and* $7\kappa/45 \le \mathbb{E}U^2 \le 2\kappa/9$. *Equality holds in* (19) *for* $Q = (Z^3 - 3Z)/\sqrt{6}$.

The scalar endpoint $\kappa = 0$ is included by continuity in the later price calculations. A nonzero normalized cubic tensor in finite dimension has $\kappa > 0$.

Write $F_{ab,c} = f_{abc}$, $A = FF^T$, $G = F^TF$, and let S be the full symmetrization of the four-index tensor A. Put

$$s = \|A\|_F^2 = \|G\|_F^2, \quad h = \|S\|_F^2, \quad V = \operatorname{tr} A^3, \quad V_4 = \operatorname{tr} A^4.$$

Symmetrization is an orthogonal projection, so $\langle A, S\rangle = h$, $0 \le h \le s$, and

$$\|S + tA\|_F^2 = t^2 s + (1 + 2t)h \qquad (t \in \mathbb{R}). \tag{20}$$

The Gaussian product formula gives

$$U = 3I_4(S) + 12I_2(G), \quad \mathbb{E}U^2 = 216h + 288s, \quad \kappa = 1296s + 1944h. \tag{21}$$

For example, the first identity follows by applying $D_iI_3(f) = 3I_2(f_i)$ and

$$I_2(g)^2 = I_4(\operatorname{Sym}(g \otimes g)) + 4I_2(g \otimes_1 g) + 2\|g\|_F^2$$

to each tensor slice. Orthogonality of Hermite degrees gives the variance. Since $\operatorname{tr} A = 1/6$, $s \le 1/36$, which also proves $\kappa \le 90$.

The sixth-cumulant identity is

$$\frac{\kappa_6}{46656} = \frac{135}{2}\operatorname{tr} S^3 + 45\operatorname{tr}(AS^2) + \frac{165}{2}\operatorname{tr}(A^2S) - 20\operatorname{tr} A^3 + \frac{135}{2}\langle S, G \otimes G\rangle. \tag{22}$$

It agrees with the real cubic identity in [6, Eq. (8.12)]. For completeness, the accompanying exact program derives it from connected Wick diagrams. For r copies of f, a loopless multigraph with multiplicities m_{ij} and degree three at each vertex has weight $(3!)^r/\prod_{i<j} m_{ij}!$. Cumulants retain precisely the connected graphs, by the exponential formula for the logarithm of the Wick generating function. There are seven labelled connected graphs in the fourth cumulant and 640 in the sixth. Expanding each term of (22), including the three pairings in S, gives the same rational coefficient in each of the six sixth-order graph classes. This checks a formal tensor identity; finite examples of tensors are not substituted for that identity.

Set

$$y = \sqrt{h/s}, \qquad a = (V/s^{3/2})^{1/3}, \qquad d = 1 + \tfrac{3}{2}y^2.$$

Then $0 \le y, a \le 1$ and

$$\kappa = 1296sd, \qquad \frac{\mathbb{E}U^2}{\kappa} = \frac{1}{9}\left(1 + \frac{1}{d}\right). \tag{23}$$

The stated variance interval follows immediately from $1 \le d \le 5/2$.

For any of the four pairs

$$(p, q) = (17/33, -1),\ (62/105, -3/4),\ (28/39, -1/2),\ (3/4, -1),$$

put

$$c_p = \frac{55/2 - 15q}{p(p-q)}, \quad c_q = \frac{55/2 - 15p}{q(q-p)}, \quad \zeta = 135/2 - c_p - c_q, \quad \beta = 20 + c_p p^3 + c_q q^3.$$

These pairs have $c_p, c_q > 0$ and $\zeta \ge 0$. Direct expansion completes the first four terms on the right of (22) as

$$c_p \operatorname{tr}(S + pA)^3 + c_q \operatorname{tr}(S + qA)^3 + \zeta \operatorname{tr} S^3 - \beta V.$$

For a real symmetric matrix B, $\operatorname{tr} B^3 \le \|B\|_F^3$. Also

$$\|\operatorname{Sym}(G \otimes G)\|_F^2 = (s^2 + 2V_4)/3, \qquad V_4 \le V^{4/3}.$$

Together with (20), these inequalities prove $\kappa_6/\kappa^{3/2} \le \mathcal{J}_{p,q}(y, a)/d^{3/2}$, where

$$\begin{aligned}\mathcal{J}_{p,q} = {} & c_p[p^2 + (1+2p)y^2]^{3/2} + c_q[q^2 + (1+2q)y^2]^{3/2} \\ & + \zeta y^3 - \beta a^3 + \frac{135}{2} y\sqrt{(1+2a^4)/3}.\end{aligned} \tag{24}$$

An additional valid bound is $\mathcal{M}(y, a)/d^{3/2}$, with

$$\begin{aligned}\mathcal{M} = {} & \frac{135}{2}y^3 + \frac{135}{2}y\sqrt{(1+2a^4)/3} + 45ay^2 - 20a^3 \\ & + \frac{165}{2}\min\left\{a^{3/2}y, y^2a^3 + ya^2\sqrt{(1-y^2)(1-a^2)}\right\}.\end{aligned} \tag{25}$$

For the first mixed bound use $\operatorname{tr}(A^2S) \le \sqrt{hV_4}$ and $V_4 \le \sqrt{s}V$. For the second, write $R = S - (h/s)A$ and apply Cauchy–Schwarz to $\langle A^2 - (V/s)A, R\rangle$, then use $V_4 \le V^{4/3}$. Finally $\operatorname{tr}(AS^2) \le V^{1/3}h$ since $\|A\|_{\mathrm{op}} \le V^{1/3}$. No positivity of S is assumed.

The exact scalar comparison, supplied in Appendix A, is

$$\frac{\min\{\mathcal{M}, \mathcal{J}_{17/33,-1}, \mathcal{J}_{62/105,-3/4}, \mathcal{J}_{28/39,-1/2}, \mathcal{J}_{3/4,-1}\}}{d^{3/2}} + \frac{211}{3d} \le \frac{194}{\sqrt{10}} + \frac{422}{15} \quad (0 \le y, a \le 1). \tag{26}$$

Its certificate consists of 84 rational rectangles and 2100 exact Bernstein coefficient checks in $\mathbb{Q}(\sqrt{10})$. Combining (23)–(26) with

$$633\left\{\frac{1}{9}(1+1/d)-\frac{7}{45}\right\}=\frac{211}{3d}-\frac{422}{15}$$

proves Theorem 4. At $Q=(Z^3-3Z)/\sqrt{6}$, $y=a=1$, $\kappa=90$, and $\kappa_6=52380=K90^{3/2}$.

6 A uniform cubic absolute-moment bound

Theorem 5. *For every normalized real third Gaussian chaos Q in every finite dimension,*

$$\mathbb{E}|Q|<\rho:=849/1000,\qquad \|P_3\|^2_{L_\infty\to L_2}\le 720801/1000000. \tag{27}$$

Put $c=\sqrt{2/\pi}$ and $R(x)=e^{x^2/2}\int_x^\infty e^{-t^2/2}\,dt$ for $x\ge 0$. The odd extension of $g(x)=-1+cR(x)$ solves $g'(x)-xg(x)=|x|-c$, with

$$g'(x)=-c(1-|x|R(|x|)).$$

The cubic Gaussian Stein identity is

$$\mathbb{E}[Qv(Q)]=\mathbb{E}[(1+U)v'(Q)],\qquad U=\tfrac{1}{3}\|DQ\|^2-1. \tag{28}$$

It follows directly from Gaussian integration by parts and the Ornstein–Uhlenbeck eigenvalue of Q. Consequently, for every odd continuously differentiable regression ψ of polynomial growth and every $\eta>0$,

$$\mathbb{E}|Q|\le c+\mathbb{E}\left[\psi'(Q)-Q\psi(Q)+\frac{(g'(Q)-\psi'(Q))^2}{2\eta}\right]+\frac{\eta}{2}\mathbb{E}U^2. \tag{29}$$

Indeed $\mathbb{E}|Q|-c=-\mathbb{E}[Ug'(Q)]$ and $\mathbb{E}[\psi'(Q)-Q\psi(Q)]=-\mathbb{E}[U\psi'(Q)]$; Young's inequality applied to $-U(g'-\psi')$ gives (29).

The eight rational rows in Appendix B supply $\eta>0$, coefficients q_0,q_2,q_4,q_6 with $q_6>0$, and a regression ψ, such that on the entire real line

$$\psi'(x)-x\psi(x)+\frac{(g'(x)-\psi'(x))^2}{2\eta}\le q_0+q_2x^2+q_4x^4+q_6x^6. \tag{30}$$

For $x\ge 0$, each regression has the explicit rational form

$$\psi(x)=v_0x+\frac{v_1x^2}{2}+\frac{v_2x^3}{3}+\frac{v_3x^4}{4}+\frac{x}{x+2}\sum_{j=0}^{12}v_{4+j}T_j\left(\frac{x-2}{x+2}\right), \tag{31}$$

extended oddly. It is continuously differentiable at zero. All v_j, all q_j, and η are rational and are supplied in the source archive as exact inputs, separately from floating discovery output.

Combining (29)–(30) with Theorem 4 gives a price affine in $\mathbb{E}U^2$. Its maximum therefore occurs at one of the two permitted variance endpoints. The resulting functions of κ are

$$\begin{aligned}p_0(\kappa)&=c+q_0+q_2+(3+\kappa)q_4+[15+15\kappa+K\kappa^{3/2}]q_6+7\eta\kappa/90,\\ p_1(\kappa)&=c+q_0+q_2+(3+\kappa)q_4+[15+15\kappa+(K-633/15)\kappa^{3/2}]q_6+\eta\kappa/9.\end{aligned} \tag{32}$$

Both are convex for $\kappa \geq 0$ because $q_6 > 0$ and $K - 633/15 > 0$. On a closed cumulant interval, four corner prices thus suffice. The eight intervals in Appendix B cover $[0, 90]$, and every directed corner enclosure is below $.848521 < .849$. This proves the first assertion of Theorem 5.

For the second assertion, normalize $P_3 f$ to a variance-one cubic Q when that projection is nonzero. For $|f| \leq 1$, $\|P_3 f\|_2 = \mathbb{E} f Q \leq \mathbb{E}|Q| < \rho$. Taking the supremum and squaring proves the projection norm bound.

Theorem 6 (A refitted uniform cubic absolute-moment bound)**.** *For every normalized real third Gaussian chaos Q in every finite dimension,*

$$\mathbb{E}|Q| < \frac{847}{1000}, \qquad \|P_3\|^2_{L_\infty \to L_2} \leq \frac{717409}{10^6}.$$

Proof. Parameterize the coupled shape by $\mathbb{E}U^2 = \kappa(7/45 + s/15)$, where $\kappa = \mathbb{E}Q^4 - 3 \in [0, 90]$ and $s \in [0, 1]$. For any complete-halfline scalar row with frozen rational parameters $(q_0, q_2, q_4, q_6, \eta)$, the Stein identity and sharp633 theorem imply

$$\mathbb{E}|Q| \leq c + q_0 + q_2 + (3 + \kappa)q_4 + [15 + 15\kappa + (K - 633s/15)\kappa^{3/2}]q_6 + \eta\kappa(7/90 + s/30),$$

where $c = \sqrt{2/\pi}$ and $K = 194/\sqrt{10}$. Eight accepted scalar rows are supplemented by two fresh rational Chebyshev rows. Their entire-halfline inequalities are verified by directed compact Taylor and exact polynomial-tail comparisons; a separate direct-Chebyshev implementation replays the new rows with 768-bit sixth-order Taylor bounds on $[0, 80]$ and independently reconstructed Fraction tails. Each row has $q_6 > 0$ and $\eta > 0$, so its price is convex in κ and affine in s on every allowed rectangle. Its validity is global in cumulant, independent of the interval where it was discovered. The exact certificate partitions $[0, 90]$ into 2780 rational intervals, selects one of the ten rows on each, and verifies all four corner prices. The largest outward rational cap is $423485406457/500000000000 = .846970812914 < .847$. An independent MPFR implementation reconstructs all 11120 selected prices and 111200 ten-row candidate checks, verifies source hashes and exact exhaustive coverage, and agrees with the bound. This proves the first assertion using only the freshly included sharp633 and Stein inputs. Normalizing a nonzero cubic projection proves the second by duality. □

Corollary 7. *Let f, g be signs, $h = (f + g)/2$, $k = (f - g)/2$, and $u = \|P_1 h\|_2$. For $T_{.72} = P_1 - (18/25)P_3$, if $u \leq 21/100$, then*

$$\langle f, T_{.72} g \rangle \leq .56307672 < 40/71. \tag{33}$$

Proof. Self-adjointness and orthogonality give

$$\langle f, T_{.72} g \rangle = \|P_1 h\|_2^2 - .72\|P_3 h\|_2^2 - \|P_1 k\|_2^2 + .72\|P_3 k\|_2^2 \leq u^2 + .72\rho^2.$$

Since $|k| \leq 1$, Theorem 5 applies. At $u = .21$ the exact right side is $441/10000 + 6487209/12500000 = .56307672$. □

The amplitude is the actual first-projection norm, without division by $\sqrt{2/\pi}$. The corollary covers a complete interval of amplitudes, but does not cover the middle range needed for (17).

With the refitted cap 847/1000, the same elementary energy proof gives $\langle f, T_{.72} g \rangle \leq u^2 + (18/25)(847/1000)^2$. It therefore also closes $0 \leq u \leq 541/2500 = .2164$, where its rational cap is $7042043/12500000 = .56336344 < 40/71$. The frozen regional certificates below retain their original, valid choice $\rho = .849$.

7 A support–source comparison

Let Z be standard Gaussian, independent of a transverse Gaussian vector W. Suppose $|h|+|k| \le 1$, set

$$u = \mathbb{E}[Zh] \ge 0, \qquad p = \mathbb{E}|k|, \qquad r = \|P_3^W \mathbb{E}_Z k\|_2.$$

The source here is the transverse cubic projection. Its independence of Z is essential to the following comparison. For $a > 0$ and $y \ge 0$ write

$$J_a(y) = \mathbb{E}_Z \max\{a|Z|, y\} = ace^{-y^2/(2a^2)} + y \operatorname{erf}(y/(\sqrt{2}a)).$$

Proposition 8. *For every $a > 0$ and $\lambda \ge 0$,*

$$au + r + \lambda p \le (1-\rho^2)J_a(\lambda) + \rho^2 J_a(\lambda + 1/\rho), \qquad \rho = 849/1000. \tag{34}$$

Proof. Set $F(y) = J_a(y+\lambda)$ and $x = 1/\rho$. Its third derivative on $[0,\infty)$ is nonpositive:

$$F'''(y) = -\frac{c(y+\lambda)}{a^3} e^{-(y+\lambda)^2/(2a^2)} \le 0.$$

Let $D = F(x) - F(0)$ and interpolate with the quadratic

$$H(y) = F(0) + \beta y + \gamma y^2, \quad \beta = 2D/x - F'(x), \quad \gamma = [xF'(x) - D]/x^2.$$

The Hermite interpolation remainder is $F'''(\xi)y(y-x)^2/6$, so $F(y) \le H(y)$ for every $y \ge 0$. Since F' is increasing and concave, the trapezoid comparison gives $D \ge x(F'(0)+F'(x))/2$, hence $\beta \ge 0$; also $D \le xF'(x)$ implies $\gamma \ge 0$. For any $Y \ge 0$ with $\mathbb{E}Y^2 = 1$ and $\mathbb{E}Y \le \rho$, therefore,

$$\mathbb{E}F(Y) \le F(0) + \beta\rho + \gamma = (1-\rho^2)F(0) + \rho^2 F(1/\rho).$$

For $r > 0$ normalize $P_3^W \mathbb{E}_Z k$ to a variance-one cubic $Q(W)$. Then $\mathbb{E}[Qk] = r$, and pointwise

$$aZh + Qk + \lambda|k| \le \max\{a|Z|, |Q| + \lambda\}.$$

Independence and Theorem 5 allow the preceding comparison with $Y = |Q|$, proving (34). If $r = 0$, choose any normalized transverse cubic: its pairing with k is zero by orthogonality. An unused Gaussian coordinate can be adjoined if necessary. □

At $\lambda = 0$, minimizing the right side of (34) minus au gives the explicit whole-amplitude source cap

$$r \le R_\rho(u) = \begin{cases} \rho, & 0 \le u \le c(1-\rho^2), \\ \rho \operatorname{erf}\sqrt{-\log v}, & c(1-\rho^2) < u < c, \\ 0, & u = c, \end{cases} \qquad v = \frac{u/c - (1-\rho^2)}{\rho^2}. \tag{35}$$

Indeed the derivative of $(1-\rho^2)ac + \rho^2 J_a(1/\rho)$ is $c[1 - \rho^2 + \rho^2 e^{-1/(2a^2\rho^2)}]$. In the middle range the minimizing value is $a = 1/(\rho\sqrt{-2\log v})$. The two endpoints follow by limits. Retaining $\lambda > 0$ preserves the actual disagreement mass in the comparison. Additional source information is still needed to close the full operator norm bound.

Lemma 9 (Physical clipping of source cells)**.** *Let $c = \sqrt{2/\pi}$, $0 < \rho < 1$, and use the support bound $r \le R_\rho(u)$ of* (35)*. Define*

$$U_\rho(r) = c\{1 - \rho^2 + \rho^2 e^{-[\mathrm{erf}^{-1}(r/\rho)]^2}\}, \qquad 0 < r < \rho,$$

with $U_\rho(0) = c$ and $U_\rho(\rho) = c(1-\rho^2)$. Every physical point in a cell $[u_-, u_+] \times [r_-, r_+]$ also satisfies

$$u \le \min\{u_+, U_\rho(r_-)\}.$$

Thus any outward rational upper bound on $U_\rho(r_-)$ may replace the cell's amplitude endpoint in all native and R3 price maxima. If this upper bound is below u_-, the cell is physically empty.

Proof. The source envelope is decreasing. For $0 < r < \rho$, solving $R_\rho(u) = r$ gives the displayed inverse; its endpoint values follow by limits. Hence $r \ge r_-$ and $r \le R_\rho(u)$ imply the asserted restriction. To certify the inverse without assuming a special-function inverse, bracket $s = \mathrm{erf}^{-1}(r_-/\rho)$ by rational numbers $s_- \le s \le s_+$ using directed monotone evaluations of erf. Since e^{-s^2} decreases for $s \ge 0$, an outward evaluation using s_- gives the required upper bound. □

7.1 Retaining the longitudinal cubic moment

The preceding comparison can also retain $v = \mathbb{E}[H_3(Z)h]$, with $H_3(z) = (z^3 - 3z)/\sqrt{6}$. Take $A > 0$, $B \ge 0$, and write

$$L(z) = Az + Bz^3 = az + dH_3(z), \qquad a = A + 3B, \quad d = \sqrt{6}B, \qquad J_{A,B}(x) = \mathbb{E}_Z \max\{|L(Z)|, |x|\}.$$

Proposition 10. *For every real x,*

$$J_{A,B}(x) \le c\left[A + 2B + \frac{x^2}{2A} - \frac{A + 6B}{24A^4}x^4 + \frac{A^2 + 20AB + 120B^2}{240A^7}x^6\right]. \tag{36}$$

If $r > 0$ and $Q = P_3^W \mathbb{E}_Z k/r$ has cumulant κ and shape $V = \mathbb{E}[(\|DQ\|^2/3 - 1)^2]$, then

$$\begin{aligned} au + dv + r \le c\Big[&q_0 + q_2 + (3 + \kappa)q_4 \\ &+ q_6\left\{15 + 15\kappa + K\kappa^{3/2} - 633\sqrt{\kappa}(V - 7\kappa/45)\right\}\Big], \end{aligned} \tag{37}$$

where (q_0, q_2, q_4, q_6) are the four coefficients in the brackets of (36)*.*

Proof. For $x \ge 0$ let $z \ge 0$ solve $x = Az + Bz^3$. Differentiating the distribution function gives

$$J'_{A,B}(x) = \Pr(|Z| \le z), \qquad J''_{A,B}(x) = \frac{ce^{-z^2/2}}{A + 3Bz^2}.$$

Moreover $J_{A,B}(0) = c(A + 2B)$ and $J'_{A,B}(0) = 0$. Set $w = z^2$, $q = x^2 = w(A + Bw)^2$, and $F(q) = J''_{A,B}(\sqrt{q})$. With $D_1 = A + Bw$ and $D_3 = A + 3Bw$,

$$F(q) = ce^{-w/2}/D_3, \qquad \frac{d}{dq} = \frac{1}{D_1 D_3}\frac{d}{dw}.$$

The recurrence

$$P_0 = 1, \qquad P_{j+1} = (P_j/2 - \partial_w P_j)D_1 D_3 + 2jBP_j D_3 + 3(2j + 1)BP_j D_1$$

gives

$$(-1)^j F^{(j)}(q) = \frac{ce^{-w/2}P_j(A,B,w)}{D_1^{2j}D_3^{2j+1}}.$$

For $j = 1, 2, 3$, the expanded numerator has respectively 5, 12 and 22 nonnegative coefficients. The explicit third numerator is given in Appendix C; hence $F'''(q) \le 0$ on the whole half-line. Taylor's formula with its integral remainder gives

$$F(q) \le \frac{c}{A} - \frac{c(A+6B)}{2A^4}q + \frac{c(A^2+20AB+120B^2)}{8A^7}q^2.$$

Integrating twice in x, using the two initial values, proves (36). Evenness covers negative x.

Pointwise $L(Z)h + Qk \le \max\{|L(Z)|, |Q|\}$, so independence and projection orthogonality give $au + dv + r \le \mathbb{E}J_{A,B}(Q)$. Apply (36) and Theorem 4. The fourth moment is exact, so its negative coefficient causes no inequality reversal; the sixth coefficient is positive. □

This comparison retains the actual longitudinal cubic moment. A global norm proof must quantify that moment jointly with the same source and profile; assigning it a favorable numerical value would not prove an operator bound.

Proposition 11 (Cubic-head contact source gate)**.** *Let $A > 0$, $B \ge 0$, $L(z) = Az + Bz^3 = az + dH_3(z)$, where $H_3(z) = (z^3 - 3z)/\sqrt{6}$, $a = A + 3B$, $d = \sqrt{6}B$, and let $J(x) = \mathbb{E}_Z \max\{|L(Z)|, |x|\}$. Set $G(s) = J(\sqrt{s})$ on $s \ge 0$. Then $G'' \le 0$, $G''' \ge 0$ and $G'''' \le 0$. For every $s_0 > 0$, the cubic Hermite interpolant P matching $G(0), G'(0), G(s_0), G'(s_0)$ satisfies*

$$J(x) \le P(x^2) = p_0 + p_2x^2 + p_4x^4 + p_6x^6 \quad (x \in \mathbb{R}), \qquad p_6 \ge 0.$$

Here $p_0 = \nu(A+2B)$, $p_2 = \nu/(2A)$, $\nu = \sqrt{2/\pi}$, and, with $D = G(s_0) - p_0 - p_2s_0$ and $E = G'(s_0) - p_2$,

$$p_4 = (3D - s_0E)/s_0^2, \qquad p_6 = (s_0E - 2D)/s_0^3.$$

Any componentwise upward rational rounding $q_j \ge p_j$ gives a global majorant with $q_6 \ge 0$. Its coefficients already contain ν.

Proof. Write $s = w(A+Bw)^2$, $D_1 = A + Bw$, $D_3 = A + 3Bw$. Since $ds/dw = D_1D_3 > 0$, Gaussian tail differentiation gives

$$G'(s) = \frac{\nu}{2}\int_0^1 \frac{e^{-t^2w(s)/2}}{A + Bw(s)}\,dt.$$

For $K_\tau(s) = e^{-\tau w(s)}/(A + Bw(s))$, $\tau \ge 0$, differentiation shows

$$K_\tau^{(j)}(s) = (-1)^j \frac{e^{-\tau w}P_j(A,B,w,\tau)}{D_1^{2j+1}D_3^{2j}}, \qquad j = 1, 2, 3.$$

Starting from $P_0 = 1$, the numerator recurrence is

$$P_{j+1} = (\tau P_j - \partial_w P_j)D_1D_3 + (2j+1)BP_jD_3 + 6jBP_jD_1.$$

For $j = 1, 2, 3$, exact expansion gives respectively 5, 12 and 22 nonnegative coefficients. Integrating the derivative signs at $\tau = t^2/2$ proves the claimed signs of G. The Hermite interpolation error is

$$G(s) - P(s) = G''''(\xi)s^2(s - s_0)^2/4! \le 0 \quad (s \ge 0),$$

including outside the interpolation interval. The leading coefficient $p_6 = G[0, 0, s_0, s_0] \ge 0$ by the sign of G'''. □

Corollary 12 (Moment price and retained longitudinal completion)**.** *Let $h = (f+g)/2$, $k = (f-g)/2$, $P_1 h = uZ$, and $v = \mathbb{E}[H_3(Z)h]$. Put $r = \|P_3^W \mathbb{E}_Z k\|_2$ and $Q = P_3^W \mathbb{E}_Z k/r$ when $r > 0$. Write $\kappa = \mathbb{E}Q^4 - 3$ and $\mathbb{E}(\|DQ\|^2/3 - 1)^2 = \kappa(7/45 + \zeta/15)$, with $0 \le \zeta \le 1$. The endpoint $\kappa = 0$ is interpreted by continuity. For a majorant from Proposition 11, the physical diamond and the coefficient-633 moment theorem give*

$$au + dv + r \le A_q(\kappa, \zeta) = q_0 + q_2 + (3+\kappa)q_4 + \left[15 + 15\kappa + \left(\frac{194}{\sqrt{10}} - \frac{633\zeta}{15}\right)\kappa^{3/2}\right] q_6.$$

No extra ν multiplies this price. For a fixed row the price is convex in κ and affine in ζ. If a certified frame completed its longitudinal penalty by $-bv^2 \le 2\beta v + \beta^2/b$, with β multiplying normalized H_3, then for any fixed $\lambda \ge 0$ its saved completed bound may be augmented by

$$\lambda\left(A_q - au - r + \frac{\beta d}{b}\right) + \frac{\lambda^2 d^2}{4b}.$$

With λ fixed throughout each price rectangle, the existing cumulant, shape, amplitude and source-chord certification can be reused. If a frame instead stores β_{raw} multiplying $z^3 - 3z$, use $\beta = \sqrt{6}\,\beta_{\mathrm{raw}}$. The same correction is then

$$\lambda\left(A_q - (A+3B)u - r + \frac{6\beta_{\mathrm{raw}} B}{b}\right) + \frac{3\lambda^2 B^2}{2b}.$$

Both the contact row and λ are chosen once for each whole cell; they cannot be optimized independently at its corner points and then treated as a single convex price.

Proof. Pointwise $L(Z)h + Qk \le \max(|L(Z)|, |Q|)$, and $\mathbb{E}Qk = r$. Take expectations, use the global majorant, and price its second, fourth and sixth moments. The sixth coefficient is nonnegative and $194/\sqrt{10} - 633/15 > 0$, proving the price properties. Before completing the longitudinal penalty, add the nonnegative quantity $\lambda(A_q - au - dv - r)$ and complete with $(\beta + \lambda d/2)^2/b$ in place of β^2/b. Subtracting the old completion gives the displayed correction. □

For the exact head $A = 26412631/10^7$, $B = 4084047/10^7$ and contact $w_0 = 189/50$, so $s_0 = w_0(A + Bw_0)^2$, one certified rational row is

$$\begin{aligned} q_0 &= 13795713289434833 0419/(5 \cdot 10^{19}),\\ q_2 &= 1510422344526876849/10^{19},\\ q_4 &= -163849393411225061/10^{20},\\ q_6 &= 23608461930109/(25 \cdot 10^{17}). \end{aligned}$$

Its uniform four-corner price on $\kappa \in [2,4]$, $\zeta \in [0,1]$ is at most $29040583649083683/10^{16}$. The exact coefficient-sign proof, rational rounding and directed corner prices are reproduced by `certify_direct_contact_majorant.py`. This source proposition alone does not assert a global operator norm bound. The scalar comparison was independently accepted by an exact rational recurrence check and a separate 768-bit MPFR replay of all 28 upward-rounded coefficients and 98 directed moment prices. A global norm conclusion still requires complete fixed-row, fixed-multiplier cell coverage.

Corollary 13 (A physical source exclusion at the remaining native bottleneck)**.** *In the notation of Proposition 11, every physical source with $u \ge 151/200$, $\kappa \in [2,4]$ and $\zeta \in [0,1]$ satisfies*

$$r \le \frac{3218564576843929081}{12500000000000000000} < \frac{257485167}{10^9} < \frac{129}{500}.$$

Thus the entire source domain $u \geq .755$, $r \geq .258$, $\kappa \in [2,4]$ is physically impossible. This exclusion is independent of the operator coefficient b and uses no longitudinal-completion multiplier.

Proof. Use the globally valid contact family at $A = 147/50$, $B = 0$, $w_0 = 11/2$, so that $s_0 = 237699/5000$. One upward-rounded rational row is

$$\begin{aligned} q_0 &= 234578060876042414629/10^{20}, \\ q_2 &= 135694653197766216 99/10^{20}, \\ q_4 &= -57073674634011611/(5 \cdot 10^{19}), \\ q_6 &= 648580884546013/10^{20} > 0. \end{aligned}$$

These coefficients already contain ν. The price for this single fixed row is convex in κ and affine in ζ. Directed evaluation of the four corners of $[2,4] \times [0,1]$ gives

$$A_q(\kappa, \zeta) \leq \frac{30964814576843929081}{12500000000000000000}.$$

The physical source gate is simply $(147/50)u + r \leq A_q$ since $d = 0$. Subtract $(147/50)(151/200)$ to obtain the asserted bound. All four rational coefficient roundings and four corner prices passed a separate 768-bit MPFR replay; the globally valid contact-family argument also applies at $B = 0$. This is a source-domain exclusion, not a complete operator norm theorem. □

8 Conditional Gaussian profiles and a mixed cubic penalty

Proposition 14 (Sharp conditional Gaussian profile and a quartic budget)**.** *Let Z be standard normal and let measurable real functions h, k satisfy $|h| + |k| \leq 1$. Put*

$$\nu = \sqrt{2/\pi}, \qquad A = \mathbb{E}[Zh], \qquad p = |\mathbb{E}k|, \qquad C = 1 - \nu^2.$$

If $p = 2\Phi(t) - 1$, $t \in [0, \infty]$, then

$$|A| \leq \nu e^{-t^2/2}. \tag{38}$$

This bound is sharp for every p. In particular,

$$A^2 + p^2 - Cp^4 \leq \nu^2. \tag{39}$$

The quartic budget improves the ellipse $A^2 + \nu^2 p^2 \leq \nu^2$ by $Cp^2(1 - p^2)$.

Proof. Write $\alpha = |k|$ and $\alpha_0 = \mathbf{1}_{\{|Z| \leq t\}}$. Since $\mathbb{E}\alpha \geq p = \mathbb{E}\alpha_0$ and $(|Z| - t)(\alpha - \alpha_0) \geq 0$ pointwise,

$$\mathbb{E}[|Z|\alpha] \geq \mathbb{E}[|Z|\alpha_0].$$

Consequently $|A| \leq \mathbb{E}[|Z|(1 - \alpha)] \leq 2\varphi(t) = \nu e^{-t^2/2}$. Equality is attained by $h = \operatorname{sgn}(Z)\mathbf{1}_{\{|Z| > t\}}$ and $k = \mathbf{1}_{\{|Z| \leq t\}}$.

For the polynomial budget set $p(t) = 2\Phi(t) - 1$, so $p' = \nu e^{-t^2/2}$ and $p'' = -tp'$. Define

$$D(t) = \nu^2 e^{-t^2} - \nu^2 + p(t)^2 - Cp(t)^4, \qquad g(t) = p(t) - 2Cp(t)^3 - tp'(t).$$

Then $D' = 2p'g$ and $g' = p'(t^2 - 6Cp^2)$. The ratio $t/p(t)$ strictly increases from $1/\nu$ to infinity: indeed $p(t) > tp'(t)$ for $t > 0$ because p' strictly decreases. Furthermore $1/\nu^2 < 6C$ and $1 - 2C > 0$; for example these follow already from $3 < \pi < 22/7$. Thus g' changes sign exactly once, from negative to positive. As $g(0) = 0$ and $\lim_{t \to \infty} g(t) = 1 - 2C > 0$, the function g has exactly one further zero. It follows that D first decreases and then increases. Its two endpoint limits are zero, so $D \leq 0$. Combining this with (38) proves (39). □

Proposition 15 (A mixed cubic completion with one scalar source)**.** *Let Z and W be independent standard Gaussian coordinates. Let f, g be odd signs and put $h = (f+g)/2$, $k = (f-g)/2$,*

$$A(W) = \mathbb{E}_Z[Zh], \qquad k_0(W) = \mathbb{E}_Z k.$$

Suppose $r = \|P_3^W k_0\|_2 > 0$ and write $Q = P_3^W k_0/r = I_3(F)$, with $6\|F\|^2 = 1$. Here Wiener integrals are normalized by $\mathbb{E}I_j(H)^2 = j!\|H\|^2$. Set

$$G = F \otimes_2 F, \quad s = \|G\|^2, \quad D = 1 + \frac{3\|\operatorname{Sym}(F \otimes_1 F)\|^2}{2s}, \quad R_2 = \frac{I_2(G)}{\sqrt{2s}}, \quad \kappa = \mathbb{E}Q^4 - 3.$$

Then $\mathbb{E}R_2 = 0$, $\mathbb{E}R_2^2 = 1$, and

$$\theta := \mathbb{E}[Q^2 R_2] = 18\sqrt{2s}, \qquad \theta^2 = \frac{\kappa}{2D}. \tag{40}$$

For $t = \mathbb{E}[AR_2]$, any $b, w > 0$ and $a, c \in \mathbb{R}$ satisfy

$$\begin{aligned} -bt^2 \le{}& w\{\nu^2 + a^2 + 2ac + c^2(3+\kappa)\} - w\mathbb{E}k_0^2 + wC\mathbb{E}k_0^4 \\ &- 2w\mathbb{E}[(a + cQ^2)A] - \frac{bwc^2\kappa}{2D(b+w)}, \end{aligned} \tag{41}$$

where $\nu^2 = 2/\pi$ and $C = 1-\nu^2$. The direction $ZR_2(W)$ is a unit cubic Hermite function orthogonal to $H_3(Z) = (Z^3 - 3Z)/\sqrt{6}$. Hence its penalty $-bt^2$ can be retained in addition to a completion using only the longitudinal cubic direction.

Proof. The Gaussian product formula yields

$$P_2(Q^2) = 18I_2(G), \qquad \kappa = 1296s + 1944\|\operatorname{Sym}(F \otimes_1 F)\|^2 = 1296sD.$$

These identities give (40). Also G is positive semidefinite, since $G_{ij} = \sum_{a,b} F_{iab}F_{jab}$. Thus R_2 is the normalized, centered quadratic form determined by this actual cubic source.

Put $\eta = a + cQ^2$. Orthogonal projection onto the unit vector R_2 gives $\mathbb{E}(A-\eta)^2 \ge (t - c\theta)^2$. Completing one scalar square gives

$$bt^2 + w(t - c\theta)^2 \ge \frac{bw}{b+w}c^2\theta^2,$$

and therefore

$$-bt^2 \le w\mathbb{E}(A-\eta)^2 - \frac{bw}{b+w}c^2\theta^2.$$

Conditionally on W, h, k satisfy $|h|+|k| = 1$, so Proposition 14 gives $A^2 \le \nu^2 - k_0^2 + Ck_0^4$. Expanding the square and using $\mathbb{E}\eta^2 = a^2 + 2ac + c^2(3+\kappa)$ proves (41). The final orthogonality claim follows from independence of Z, W and $\mathbb{E}R_2 = 0$. □

Insertion into a local scalar criterion. If the original conditional potential contains $2\ell A$ and local moments $\mathbf{k} = (k_0, k_1, k_2, k_3)$, formula (41) replaces

$$\ell \quad \text{by} \quad \ell - w(a + cq^2)$$

at scalar source value q, adds the local polynomial $-wk_0^2 + wCk_0^4$, and adds only the explicit outer price

$$w\{\nu^2 + a^2 + 2ac + c^2(3+\kappa)\} - \frac{bwc^2\kappa}{2D(b+w)}.$$

It introduces no additional source or local moment coordinate. This insertion is an exact inequality, not a global norm certificate: the modified local potential and its source tail must still be bounded on their complete domains. In particular the quadratic dependence on q requires a new chord allowance. If the unmodified local potential is a supremum of affine functions of q, then its modified physical supremum Ψ obeys, for $q \in [q_0, q_1]$,

$$\Psi(q) \le \mathrm{Chord}_{q_0,q_1} \Psi(q) + 2w|c|\nu(q-q_0)(q_1-q).$$

Indeed adding $2w|c|\nu q^2$ makes every candidate convex, because $|A| \le \nu$. Any pre-existing curvature allowance must also be retained.

9 Continuous regional norm certificates

We next convert local inequalities into bounds on complete intervals of the agreement amplitude. All source and cumulant parameters are covered continuously; finite sampling is used only to choose the rational inputs.

Throughout this section the longitudinal Hermites are orthonormal:

$$H_0 = 1, \quad H_1 = Z, \quad H_2 = (Z^2-1)/\sqrt{2}, \quad H_3 = (Z^3-3Z)/\sqrt{6}.$$

Section 10 explicitly switches to the raw Hermite convention.

Proposition 16 (A transverse Gram comparison)**.** *Let $b > 0$, let f, g be odd signs, put $h = (f+g)/2$, $k = (f-g)/2$, and rotate so that $P_1 h = uZ$. In the transverse coordinates W, set $k_j(W) = \mathbb{E}_Z[H_j(Z)k]$, $h_j(W) = \mathbb{E}_Z[H_j(Z)h]$, and $r = \|P_3^W k_0\|_2$. Suppose the even block E on coordinates $(0,2)$ and odd block R on coordinates $(1,3)$ satisfy*

$$E \succeq 0, \quad E - \mathrm{diag}(-1,b) \succeq 0, \quad R \succeq 0, \quad R - \mathrm{diag}(-1,b) \succeq 0, \quad R - \mathrm{diag}(b,0) \succeq 0, \qquad E_{22} > 0.$$

Let M be the corresponding four-coordinate matrix. For arbitrary real $\ell, \beta, \delta, \omega$, define

$$s = (\delta, 0, -\omega, 0), \qquad \xi = \frac{(\omega - E_{02})^2}{E_{22}} - E_{00} + b - 2\delta.$$

Let $\mathcal{P}(t)$ be the supremum, over conditional functions satisfying $|h| + |k| \le 1$, of

$$\mathbf{k}^T M \mathbf{k} + 2t\, s^T \mathbf{k} + 2\ell h_1 + 2\beta h_3 + \beta^2/b, \qquad \mathbf{k} = (k_0, k_1, k_2, k_3)^T.$$

For $r > 0$, with $Q = P_3^W k_0 / r$, one has

$$\langle f, T_b g \rangle \le u^2 - 2\ell u + \mathbb{E}\mathcal{P}(rQ) + \xi r^2. \tag{42}$$

For $r = 0$, replace the expectation by $\mathcal{P}(0)$.

Proof. Retain $-b(\mathbb{E}h_3)^2$ from the agreement penalty and use $-bx^2 \le 2\beta x + \beta^2/b$. Expand the disagreement function by longitudinal and transverse Hermite degree. In transverse degree one, the even block must dominate $\mathrm{diag}(-1,b)$; in transverse degrees zero and two, the odd block must dominate the two displayed diagonal matrices. In the source direction Q, write $a = \mathbb{E}[Qk_2]$. The even quadratic and source terms are

$$E_{22}a^2 + 2(E_{02} - \omega)ra + (E_{00} + 2\delta + \xi)r^2 \ge br^2,$$

by their exact Schur completion. The component of $P_3^W k_0$ orthogonal to Q is zero by definition. All remaining transverse components are dominated by the positive blocks. Finally insert $2\ell\mathbb{E}Zh - 2\ell u = 0$. These comparisons give (42) after conditional maximization. □

For completeness, the finite local comparison is particularly simple. If the actual conditional moment vector lies in a box centered at c with half widths w, exact quadratic completion and the diamond constraint give the entire box bound

$$\mathcal{L}(t;h,k) \le 2\mathbb{E}_Z \max\{|(Mc+ts)\cdot H|, |\ell H_1 + \beta H_3|\} - c^T Mc + \beta^2/b + w^T|M|w, \tag{43}$$

where $\mathcal{L}$ denotes the objective defining $\mathcal{P}$ and $|M|$ is entrywise absolute value. This is a bound for actual moments in that box, with no stationary-point assumption. Bessel's inequality and $|k_j| \le \mathbb{E}|H_j|$ restrict the moments to

$$\sum_{j=0}^{3} k_j^2 \le 1, \qquad (|k_0|,|k_1|,|k_2|,|k_3|) \le (1, 4/5, 7/10, 5/8).$$

The expectation in (43) is the sum of two Gaussian absolute-cubic integrals, since $\max(|A|,|B|) = (|A+B| + |A-B|)/2$. Certified real-root isolation and Gaussian antiderivatives evaluate these integrals with outward rounding.

The potential $\mathcal{P}$ is even and convex in t. Its certified nodal heights therefore give upper chords on every source interval; a tail slope at least $2\|s\|$ gives an affine extension to infinity. Denote this even convex upper function by F. To price $\mathbb{E}F(rQ)$, solve $g_F'(x) - xg_F(x) = F(rx) - \mathbb{E}F(rN)$ for a standard normal N. For any $\lambda \ge 0$, any odd regression ψ, and $\eta > 0$, the same Stein calculation as in Section 6 gives

$$\mathbb{E}F(rQ) \le \mathbb{E}F(rN) + \lambda(\rho - c) + \mathbb{E}\left[\psi'(Q) - Q\psi(Q) + J_\eta(g_F'(Q) - \lambda g'(Q) - \psi'(Q))\right] + \frac{\eta}{2}\mathbb{E}U^2, \tag{44}$$

$$J_\eta(x) = \begin{cases} x^2/(2\eta), & x \le \eta, \\ x - \eta/2, & x \ge \eta. \end{cases}$$

Here $c = \sqrt{2/\pi}$, g is the absolute-value Stein solution above, and $\rho = 849/1000$. The clipped quadratic follows by maximizing $-vx - \eta v^2/2$ over $v \ge -1$, since $U \ge -1$. Smooth approximation justifies the finitely many target knots.

A whole-line sextic majorant for the bracket in (44), with nonnegative sextic coefficient, has a price convex in κ and affine in the shared variance shape. Consequently its four endpoint comparisons cover each complete cumulant–shape rectangle. Source interpolation adds only

$$(-\xi)_+ \max_i (r_{i+1} - r_i)^2/4$$

to the endpoint prices, because $\mathbb{E}F(rQ)$ is convex in r. Checking the two endpoints of $u^2 - 2\ell u$ covers an entire amplitude interval. These facts explain the continuous scope of the following certificates.

Theorem 17 (Complete regional amplitude bounds)**.** *For odd Gaussian signs f, g, put $u = \|P_1((f+g)/2)\|_2$. In every finite Gaussian dimension,*

$$\langle f, T_{.72} g\rangle < \begin{cases} 563367/10^6, & 0 \le u \le 341/500, \\ 563334/10^6, & 77/100 \le u \le \sqrt{2/\pi}. \end{cases} \qquad < \frac{40}{71}.$$

Proof. The seven frozen profiles in Appendix D cover the intervals $[.14, .42]$, $[.30, .485]$, $[.40, .54]$, $[.50, .638]$, $[.60, .662]$, $[.662, .682]$, and $[.77, .797885]$. Each satisfies Proposition 16, complete local

moment-box comparisons, and complete scalar inequalities of (44). Directed prices and exact convex endpoint comparisons cover every source amplitude, cumulant and shared shape in each region. Their first six rational caps are all at most $563367/10^6$; the seventh cap is $563334/10^6$. For $u \le .14$, the energy comparison gives $u^2 + .72\rho^2 \le .53857672$. The intervals therefore cover every $u \le .682$. Finally $u \le \mathbb{E}|Z| = \sqrt{2/\pi} < .797885$, so the last interval contains every feasible $u \ge .77$. □

The accepted broad profile uses a source grid ending exactly at ρ, which is valid at every agreement amplitude by projection norm duality. Hence its original discovery interval imposes no restriction on its larger accepted interval. The other extensions retain their verified outward source endpoint. This accounts for the continuous scope of the theorem. The remaining entire amplitude range is $.682 < u < .77$; a local source-parameter subregion inside that range does not complete its full norm cover.

10 Retaining the actual cubic projection

Let $T_b = P_1 - bP_3$, $b > 0$, on real standard Gaussian space, where P_j is the orthogonal projection onto total Hermite degree j. For odd signs f, g put $h = (f+g)/2$, $k = (f-g)/2$. Then (h, k) takes the four values $(\pm 1, 0), (0, \pm 1)$. Rotate the Gaussian coordinates as (Z, W) so that $P_1 h = uZ$, with Z independent of W. Use the raw longitudinal Hermites

$$H_0 = 1, \quad H_1 = Z, \quad H_2 = Z^2 - 1, \quad H_3 = Z^3 - 3Z, \qquad m_j(W) = \mathbb{E}_Z[H_j k].$$

If $r = \|P_3^W m_0\|_2 > 0$, define the normalized cubic source $Q = P_3^W m_0 / r$. Thus $\mathbb{E}Q^2 = 1$ and $\mathbb{E}[Q m_0] = r$. All remaining projections act on W.

Proposition 18 (An exact residual of the physical cubic source)**.** *Put*

$$\kappa = \mathbb{E}Q^4 - 3, \quad c_6 = \mathbb{E}Q^6 - 15\mathbb{E}Q^4 + 30, \quad S(q) = q^3 - 3q,$$

$$\mathcal{R} = Q^3 - (3+\kappa)Q - P_1^W(Q^3), \quad \Delta = \mathcal{R} - S(Q), \quad D = \kappa^2 + \|P_1^W(Q^3)\|_2^2.$$

The residual $\mathcal{R}$ is odd, orthogonal to first chaos and to Q, and

$$\mathbb{E}\mathcal{R}^2 = 6 + 9\kappa + c_6 - D, \quad \mathbb{E}\Delta^2 = D, \quad \mathbb{E}[Q\Delta] = -\kappa, \quad \mathbb{E}[S(Q)\Delta] = -D. \tag{45}$$

Proof. Since Q has pure degree three, $P_1^W(Q^3)$ is orthogonal to Q. The coefficient removed from Q^3 along the unit vector Q is $\mathbb{E}Q^4 = 3 + \kappa$. Also $\Delta = -\kappa Q - P_1^W(Q^3)$. Orthogonal expansion of these two expressions proves all four identities. □

Write $Q = I_3(F)$ and put $s_F = \|F \otimes_2 F\|_F^2$ and $z = 1296 s_F / \kappa \in [2/5, 1]$, as in the tensor shape comparison of Section 5. The residual parameter has the dimension-free bounds

$$\kappa^2 \le D \le \kappa^2 + \frac{9}{4}(\kappa z)^{3/2}. \tag{46}$$

Indeed Gaussian differentiation and the product formula give $\mathbb{E}[X_i Q^3] = 324\langle F_i, F \otimes_2 F\rangle$. The flattening map of F has squared operator norm at most $\sqrt{s_F}$, so the sum of the squared inner products is at most $s_F^{3/2}$. Multiplication by 324^2 proves (46).

Proposition 19 (An inverse-free physical dual criterion)**.** *Fix $p = (p_0, p_2)$ and a symmetric matrix $R = \left(\begin{smallmatrix} R_{11} & R_{13} \\ R_{13} & R_{33} \end{smallmatrix}\right)$ such that*

$$p_2^2 \geq \frac{b}{2}(1 + p_0^2), \qquad R \succeq 0, \quad R - \mathrm{diag}(-1, b/6) \succeq 0, \quad R - \mathrm{diag}(b, 0) \succeq 0. \tag{47}$$

Choose real $\ell, \beta, \gamma, y, \zeta, c, d$ and $G > 0$. Write

$$\sigma^2 = p_0^2 + 2p_2^2, \quad \rho = 1 + \zeta^2/G, \quad \mathfrak{g}(q) = cq + dS(q), \quad a(q) = \zeta[-S(q) + \mathfrak{g}(q)/(2G)].$$

For $v = (s, v_1, v_3)$ define

$$\begin{aligned} L(z) &= \ell z + \beta H_3(z), \\ B(q, z; v) &= q\gamma + q(y - p_0 p_2 r) H_2(z) + [a(q) + \rho s][p_0 + p_2 H_2(z)] \\ &\quad + (R_{11} v_1 + R_{13} v_3) z + (R_{13} v_1 + R_{33} v_3) H_3(z), \end{aligned}$$

and the physical dual ellipsoid

$$\mathcal{E}_p = \{(s, v_1, v_3) : s^2/\sigma^2 + v_1^2 + v_3^2/6 \leq 1\}.$$

Put

$$\Phi(q) = \sup_{v \in \mathcal{E}_p} \left\{ 2\mathbb{E}_Z \max(|L(Z)|, |B(q, Z; v)|) - \rho s^2 - R_{11} v_1^2 - 2R_{13} v_1 v_3 - R_{33} v_3^2 + \frac{\mathfrak{g}(q)^2}{4G} \right\}. \tag{48}$$

Then

$$\begin{aligned} \langle T_b f, g \rangle \leq u^2 - 2\ell u + \frac{6\beta^2}{b} + (b - p_0^2) r^2 - 2\gamma r + \frac{y^2}{p_2^2} + \mathbb{E}\Phi(Q) \\ + \zeta^2(6 + 9\kappa + c_6) - c\kappa + (G - d - \zeta^2) D. \end{aligned} \tag{49}$$

The ellipsoid in (48) *may be restricted to $v_1 \geq 0$ without changing the supremum, and $\Phi(-q) = \Phi(q)$.*

Proof. Let $V = p_0 m_0 + p_2 m_2$ and $t = \mathbb{E}[Q m_2]$. Let Π project onto odd transverse degrees at least three, orthogonally to Q. Hermite orthogonality and (47) give

$$\begin{aligned} -\|P_1 k\|_2^2 + b\|P_3 k\|_2^2 \leq \mathbb{E}V^2 + \mathbb{E}(R_{11} m_1^2 + 2R_{13} m_1 m_3 + R_{33} m_3^2) \\ + (b - p_0^2) r^2 - 2p_0 p_2 r t - p_2^2 t^2 - \|\Pi V\|_2^2. \end{aligned}$$

Indeed the relevant raw-Hermite blocks are $\mathrm{diag}(-1, b/2)$ on (m_0, m_2) in transverse degree one, $\mathrm{diag}(-1, b/6)$ on (m_1, m_3) in degree zero, and $\mathrm{diag}(b, 0)$ on that latter pair in degree two. The first gate is exactly the first block inequality for pp^T. In even longitudinal/transverse degree three, $P_3^W m_0 = rQ$; subtracting the Q component of V leaves precisely the indicated unused loss. Higher odd transverse degrees supply the remaining part of $\|\Pi V\|_2^2$.

Now $\mathcal{R}$ lies in the range of Π, so

$$-\|\Pi V\|_2^2 \leq -2\zeta \mathbb{E}[\mathcal{R} V] + \zeta^2 \mathbb{E}\mathcal{R}^2.$$

The pointwise square inequality

$$-2\zeta \Delta V \leq \mathfrak{g}(Q)\Delta + G\Delta^2 + (\zeta V + \mathfrak{g}(Q)/2)^2/G$$

and (45) yield the last line of (49); the remaining local quadratic is ρV^2 with linear coefficient $2a(Q)V$ and constant $\mathfrak{g}(Q)^2/(4G)$. Complete $-p_2^2t^2 \le 2yt + y^2/p_2^2$, and insert the exact zeros $2\gamma\mathbb{E}[Qm_0] - 2\gamma r$ and $2\ell\mathbb{E}[Zh] - 2\ell u$. Finally $-b\|P_3h\|_2^2 \le -(b/6)(\mathbb{E}[H_3h])^2 \le 2\beta\mathbb{E}[H_3h] + 6\beta^2/b$.

For each fixed W, Bessel's inequality gives $m_0^2 + m_1^2 + m_2^2/2 + m_3^2/6 \le \mathbb{E}_Z k^2 \le 1$. Hence $(V, m_1, m_3) \in \mathcal{E}_p$ by Cauchy–Schwarz in the even block. For a positive quadratic form A, the identity

$$x^TAx = \sup_v\{2v^TAx - v^TAv\}$$

holds also when A is singular, with equality at $v = x$. Apply it to $\mathrm{diag}(\rho, R)$ and interchange the two suprema. Maximizing the resulting linear functional over the four pointwise possibilities for (h, k) gives $2\max(|L|, |B|)$. Restricting the dual supremum to $\mathcal{E}_p$ loses no value because it contains every physical moment vector. Conversely every dual vector gives at most the physical supremum by the same quadratic identity. This proves (48) without a matrix inverse.

Under $z \mapsto -z$, the simultaneous reflection $(v_1, v_3) \mapsto (-v_1, -v_3)$ preserves the value and the ellipsoid. Thus $v_1 \ge 0$ suffices. Under $(q, s) \mapsto (-q, -s)$ the even part of B changes sign; reflecting z and changing the overall sign of B restores the same absolute value. This proves the evenness in q. □

Proposition 20 (A full even-block criterion)**.** *Retain the physical residual-source notation, including $m_j = \mathbb{E}_Z[H_jk]$. Let $E = (E_{ij})_{i,j\in\{0,2\}}$, $R = (R_{ij})_{i,j\in\{1,3\}}$ be symmetric matrices and $p = (p_0, p_2)^T$ satisfying*

$$\begin{aligned} &E \succeq 0, && E - \mathrm{diag}(-1, b/2) \succeq 0, && E - pp^T \succeq 0,\\ &R \succeq 0, && R - \mathrm{diag}(-1, b/6) \succeq 0, && R - \mathrm{diag}(b, 0) \succeq 0. \end{aligned}$$

In particular $E_{22} \ge b/2 > 0$. For arbitrary real $\ell, \beta, \gamma, y, \zeta, c, d$ and $G > 0$, put

$$\begin{gathered} S(q) = q^3 - 3q, \quad \mathfrak{g}(q) = cq + dS(q),\\ a(q) = \zeta[-S(q) + \mathfrak{g}(q)/(2G)], \quad L(z) = \ell z + \beta H_3(z). \end{gathered}$$

With the coordinate order $(0, 1, 2, 3)$, let N have even block $E + (\zeta^2/G)pp^T$ and odd block R, and set

$$b_0(q) = (q\gamma + a(q)p_0,\ 0,\ q(y - E_{02}r) + a(q)p_2,\ 0)^T, \quad H(z) = (H_0, H_1, H_2, H_3)^T.$$

Define the physical ellipsoid and local dual value by

$$\begin{aligned} \mathcal{E} &= \{v : v_0^2 + v_1^2 + v_2^2/2 + v_3^2/6 \le 1\},\\ \Phi_E(q) &= \sup_{v\in\mathcal{E}} \left\{ 2\mathbb{E}_Z \max\big(|L(Z)|, |(b_0(q) + Nv)^TH(Z)|\big) - v^TNv + \frac{\mathfrak{g}(q)^2}{4G} \right\}. \end{aligned}$$

Then

$$\begin{aligned} \langle T_bf, g\rangle \le u^2 - 2\ell u + \frac{6\beta^2}{b} + (b - E_{00})r^2 - 2\gamma r + \frac{y^2}{E_{22}} + \mathbb{E}\Phi_E(Q)\\ + \zeta^2(6 + 9\kappa + c_6) - c\kappa + (G - d - \zeta^2)D. \end{aligned}$$

The supremum may be restricted to $v_1 \ge 0$, and Φ_E is even.

Proof. Put $t = \mathbb{E}[Qm_2]$, $V = p_0 m_0 + p_2 m_2$, and let Π project onto odd transverse degrees at least three, orthogonally to Q. Hermite orthogonality gives

$$-\|P_1 k\|_2^2 + b\|P_3 k\|_2^2 \le \mathbb{E}(m_0, m_2) E (m_0, m_2)^T + \mathbb{E}(m_1, m_3) R (m_1, m_3)^T + (b - E_{00}) r^2 - 2E_{02} rt - E_{22} t^2 - \|\Pi V\|_2^2.$$

To verify this inequality, the transverse degree-one even block is $\mathrm{diag}(-1, b/2)$, while the degree-zero and degree-two odd blocks are $\mathrm{diag}(-1, b/6)$ and $\mathrm{diag}(b, 0)$. The corresponding gates cover them. In transverse degree three, $P_3^W m_0 = rQ$ exactly. Subtracting the Q-component of the E quadratic leaves a nonnegative multiple $E_{22}\|P^W_{3,Q^\perp} m_2\|_2^2$. This covers the same component of $\|\Pi V\|_2^2$, since $E_{22} \ge p_2^2$. In every higher odd transverse degree, $E - pp^T \succeq 0$ covers the unused loss. The other odd-block degrees use $R \succeq 0$.

Apply the residual square completion to V exactly as before: $-\|\Pi V\|_2^2 \le -2\zeta \mathbb{E}[\mathcal{R} V] + \zeta^2 \mathbb{E}\mathcal{R}^2$, followed by the completion of $-2\zeta \Delta V$. This adds only $(\zeta^2/G) pp^T$ to the even block and gives the displayed D, c_6, κ terms. Complete $-E_{22} t^2 \le 2yt + y^2/E_{22}$ and the longitudinal agreement penalty, and insert the exact linear and source identities. For each transverse point, Bessel's inequality places the physical four-moment vector in $\mathcal{E}$. The identity $x^T N x = \sup_v (2v^T N x - v^T N v)$, valid also for singular positive semidefinite N, and the four pointwise values of (h, k) now give precisely Φ_E. The reflections $z \mapsto -z$ and $(q, v_0, v_2) \mapsto (-q, -v_0, -v_2)$ give the final symmetry claims. □

Transport and a necessary concavity allowance. Set $y(r) = y_0 + E_{02}(r - r_0)$ and $v_y = y_0 - E_{02} r_0$. This freezes b_0 and hence the whole local certificate, while

$$(b - E_{00}) r^2 - 2\gamma r + \frac{y(r)^2}{E_{22}} = a_r r^2 - 2\left(\gamma - \frac{E_{02} v_y}{E_{22}}\right) r + \frac{v_y^2}{E_{22}}, \qquad a_r = b - E_{00} + \frac{E_{02}^2}{E_{22}}.$$

Unlike the rank-one case, a_r can be negative. On a source cell $r \in [r_-, r_+]$, endpoint interpolation remains an upper bound after adding $\max(0, -a_r)(r_+ - r_-)^2/4$. No compact local certificate, or fixed-source price, alone proves a global operator norm bound.

Proposition 21 (Jensen reductions with a rank local oracle)**.** *Retain the raw-Hermite rank residual-source notation $p = (p_0, p_2)$, L, B, $\mathcal{E}_p$, Φ, $b > 0$ and R. Let $0 \le \omega < 1$, $\nu^2 = 2/\pi$, and put $C_0 = 1 - \nu^2$. Use the exact source transport*

$$y(r) = y_0 + p_0 p_2 (r - r_0), \qquad v_y = y_0 - p_0 p_2 r_0, \qquad \eta = \gamma - p_0 v_y / p_2.$$

Write $\mathcal{C}$ for the usual residual price, including $6\beta^2/b$ and v_y^2/p_2^2, but excluding the amplitude and source quadratics and $\mathbb{E}\Phi(Q)$. Assume the three usual odd-block spectral gates for R.

For the ellipse reduction choose any $v_{\mathrm{e}} \ge \nu^2$ and assume only

$$p_2^2 \ge \frac{b}{2}\left(1 + \frac{p_0^2}{1 + v_{\mathrm{e}}\omega}\right). \tag{50}$$

Then the ordinary rank local oracle, with constant longitudinal head, gives

$$\langle T_b f, g\rangle \le (1 - \omega) u^2 - 2\ell u + (b - v_{\mathrm{e}}\omega) r^2 - 2\eta r + \mathcal{C} + v_{\mathrm{e}}\omega + \mathbb{E}\Phi(Q). \tag{51}$$

The stronger old gate $p_2^2 \ge (b/2)(1 + p_0^2)$ is not required.

For the quartic reduction replace the even gate by

$$p_2^2 \ge \frac{b}{2}\left(1 + \frac{p_0^2}{1 + \omega}\right). \tag{52}$$

Define the modified local oracle by adding one independent bounded dual variable $\tau \in [-1,1]$ to the rank oracle: replace $B(q,Z;v)$ inside $2\mathbb{E}_Z \max(|L|,|B|)$ by $B(q,Z;v) + 2\omega C_0\tau^3$, subtract $3\omega C_0\tau^4$ outside this expectation, and maximize over both $v \in \mathcal{E}_p$ and $\tau \in [-1,1]$. Denote the result by $\Phi_\omega(q)$. Then

$$\langle T_b f, g\rangle \le (1-\omega)u^2 - 2\ell u + (b-\omega)r^2 - 2\eta r + \mathcal{C} + \omega\nu^2 + \mathbb{E}\Phi_\omega(Q), \tag{53}$$

and $0 \le \Phi_\omega(q) - \Phi(q) \le \omega C_0$. Neither extension introduces a quadratic dependence of the longitudinal head on q. Existing residual curvature allowances remain necessary; the quartic extension adds no new such allowance and its coarse tail increases by at most ωC_0.

Proof. Set $A(W) = \mathbb{E}_Z[Zh]$ and $m_0(W) = \mathbb{E}_Z k$. Alignment of $P_1 h = uZ$ gives $\mathbb{E}A = u$, and Jensen gives $\mathbb{E}A^2 \ge u^2$. The conditional profile budget proves

$$A^2 + \nu^2 m_0^2 \le \nu^2, \qquad A^2 + m_0^2 - C_0 m_0^4 \le \nu^2.$$

Thus the integrated quantities

$$v_{\mathrm{e}} - u^2 - v_{\mathrm{e}}\mathbb{E}m_0^2, \qquad \nu^2 - u^2 - \mathbb{E}m_0^2 + C_0\mathbb{E}m_0^4$$

are nonnegative, using $v_{\mathrm{e}} \ge \nu^2$ for the first.

Apply the full even-block criterion with $E = pp^T + ae_0e_0^T$, taking $a = v_{\mathrm{e}}\omega$ or $a = \omega$. The gates $E \succeq 0$ and $E - pp^T \succeq 0$ are automatic. The remaining gate is equivalent to

$$\det(E - \mathrm{diag}(-1, b/2)) = (1+a)p_2^2 - \frac{b}{2}(1 + a + p_0^2) \ge 0,$$

which is exactly the respective displayed even gate. In particular $p_2 \ne 0$. Add ω times the corresponding nonnegative integrated budget to the full criterion. Its negative m_0^2 term cancels the extra local even-block entry exactly, so the effective local quadratic remains the original rank quadratic. The extra outer terms are $\omega(v_{\mathrm{e}} - u^2 - r^2 v_{\mathrm{e}})$ in the ellipse case and $\omega(\nu^2 - u^2 - r^2)$ in the quartic case. Exact $y(r)$ transport then gives the two stated prices.

In the quartic case the additional local term is $\omega C_0 m_0^4$. Convexity of x^4 gives, for $|x| \le 1$,

$$x^4 = \sup_{|\tau|\le 1} (4\tau^3 x - 3\tau^4),$$

with equality at $\tau = x$. Since $m_0 = \mathbb{E}_Z k$, the linear part shifts the support field by $2\omega C_0\tau^3$, proving the stated local formula by the same PSD duality as for the rank oracle. The actual physical moment vector and $\tau = m_0$ are admissible, and the two dual suprema may be interchanged with the physical supremum. The added physical term lies between zero and ωC_0, proving the local gap and tail statements. The extra dual terms do not depend on q, and $1-\omega > 0$ preserves the positive amplitude curvature needed for exact polygon pricing. □

These are analytic sufficient criteria. Any newly fitted parameters, including ones violating the stronger old rank gate, require validation of the true even-block gates and a complete local scalar certificate before yielding an operator norm bound.

Proposition 22 (A frozen-centering scalar majorant)**.** *Let $U = \|\nabla_W Q\|^2/3 - 1$ and $\mathcal{V} = \mathbb{E}U^2$. Fix any $\theta \in \mathbb{R}$, $\eta > 0$, real a_0, a_2, and a continuously differentiable test function φ such that φ and φ' have at most polynomial growth. Set*

$$e(q) = \varphi'(q) - a_0 - a_2(q^2-1), \qquad \lambda(q) = -1 - \theta(q^2-1),$$

$$P(q)=\begin{cases} e(q)^2/(2\eta), & e(q)\ge \eta\lambda(q),\\ \lambda(q)e(q)-\eta\lambda(q)^2/2, & e(q)<\eta\lambda(q).\end{cases}$$

If, for every real q,

$$\Phi(q)\le A_0+A_2q^2+A_4q^4+A_6q^6+q\varphi(q)-[1+\theta(q^2-1)]\varphi'(q)-P(q), \tag{54}$$

then

$$\begin{aligned}\mathbb{E}\Phi(Q)\le{}& A_0+A_2+(3+\kappa)A_4+(15+15\kappa+c_6)A_6\\ &+\frac{\eta}{2}[\mathcal{V}-2\theta\kappa/3+\theta^2(\kappa+2)]+a_2[\kappa/3-\theta(\kappa+2)].\end{aligned} \tag{55}$$

In particular one may freeze $\theta=\kappa_0/[3(\kappa_0+2)]$ throughout a whole cumulant interval; the covariance correction in (55) *keeps its displayed positive sign. When φ is odd, the proposed majorant is even, so it suffices to verify* (54) *for $q\ge 0$. The same proof applies to the full-even-block potential Φ_E after replacing Φ by Φ_E.*

Proof. Since Q has pure degree three, Gaussian integration by parts gives $\mathbb{E}[Q\varphi(Q)]=\mathbb{E}[(1+U)\varphi'(Q)]$. It also gives $\mathbb{E}U=0$ and $\mathbb{E}[U(Q^2-1)]=\kappa/3$; the latter follows by using $\varphi(q)=q^3$. Put $U_\theta=U-\theta(Q^2-1)$. Then $U_\theta\ge\lambda(Q)$, its variance is $\mathcal{V}-2\theta\kappa/3+\theta^2(\kappa+2)$, and $\mathbb{E}[U_\theta(Q^2-1)]=\kappa/3-\theta(\kappa+2)$. By construction $P(q)=\sup_{v\ge\lambda(q)}[ve(q)-\eta v^2/2]$. Therefore $U_\theta e(Q)-P(Q)\le\eta U_\theta^2/2$. Take expectations in (54) and apply these identities. This proves (55). □

Exact transport of the source amplitude. Freeze a reference r_0 and y_0, and set

$$y(r)=y_0+p_0p_2(r-r_0),\qquad v_y=y_0/p_2-p_0r_0.$$

Then $y(r)-p_0p_2r=y_0-p_0p_2r_0$, so the entire function Φ is unchanged for every r. Its outer quadratic simplifies exactly to

$$(b-p_0^2)r^2-2\gamma r+y(r)^2/p_2^2=br^2-2(\gamma-p_0v_y)r+v_y^2. \tag{56}$$

Consequently no local amplitude error is needed for this transport; both the u and r outer quadratics are convex and are bounded on rectangles by their vertex values. The source moments and cumulants must still be covered separately. A proposed scalar majorant becomes a norm certificate only after it bounds (48) for the whole real q line and its outer expectation is bounded for every admissible physical source. No finite grid maximization establishes those requirements. The case $r=0$ can be handled separately by a source-free criterion, without normalizing a zero projection.

Proposition 23 (A certified cubic-source rectangle)**.** *For $b=18/25$, every odd Gaussian sign pair whose physical parameters satisfy*

$$u\in[7549/10000,7551/10000],\qquad r\in[2574/10000,2577/10000],\qquad \kappa\in[51/100,55/100]$$

obeys

$$\langle(P_1-bP_3)f,g\rangle\le\frac{28167}{50000}=0.56334<\frac{40}{71}.$$

The bound covers every admissible cubic tensor shape and every Gaussian dimension.

Certificate proof. The exact rational profile is frozen in `frozen_R3_profile_exact.json`. The scalar majorant uses reserves $2\cdot10^{-5}$ in its constant coefficient and 10^{-6} in its degree-six coefficient. Exact rational Bernstein derivative bounds and directed 256-bit MPFR arithmetic certify the whole interval $q\in[0,8]$: the completed partition has 6751 nodes and 3376 accepted leaves. A fresh independent traversal rechecked all accepted leaves and the physical-center quadratic radii, with stable source hashes. The independently replayed $[8,64]$ cover and exact nonnegative-coefficient tail proof for $q\ge64$ remain valid after increasing the constant reserve. Evenness covers negative q.

Use the fresh sixth-cumulant comparison and the exact cubic projection bound

$$c_6\le\left[\frac{194}{\sqrt{10}}-\frac{633}{9}(z-2/5)\right]\kappa^{3/2},\qquad D\le\kappa^2+\frac{9}{4}(\kappa z)^{3/2},\quad z\in[2/5,1].$$

In the outer price their respective coefficients are nonnegative. After the exact $y(r)$ transport, the price is separately convex in u,r,κ,z; thus its 16 rectangle vertices suffice. Independent directed evaluation gives the upper bound $0.563333433603497\ldots<28167/50000$. □

This certifies the displayed rectangle of physical parameters. It is not a global bound for $\|P_1-(18/25)P_3\|_{\infty\to1}$.

Proposition 24 (Certified moving bands of cubic-source parameters)**.** *Let $b=18/25$ and use the physical parameters u,r,κ of an odd Gaussian sign pair. Define*

$$t=r-\frac{5151}{20000}+3\left(u-\frac{151}{200}\right).$$

In each of the following three cases, the pairing is bounded by the listed cap, in every Gaussian dimension and for every admissible dimensionless cubic tensor shape $z\in[2/5,1]$:

u	$\lvert t\rvert$ *at most*	κ	$\langle(P_1-bP_3)f,g\rangle$ *at most*
$[.7538,.7562]$	.00025	$[.3,.8]$	28168/50000
$[.75,.76]$	.001	$[.3,.8]$	451/800
$[.7505,.7595]$	.001	$[.15,1.1]$	451/800

All finite decimals in the table are exact rational numbers. The first cap is $0.56336<40/71$; the other two are 0.56375.

Certificate proof. Use the accepted whole-line scalar majorant from the exact frozen profile, with constant reserve $1/50000$ and sixth-degree reserve $1/1000000$. The transport $y(r)=y_0+p_0p_2(r-r_0)$ leaves this entire local majorant unchanged. Its outer amplitude part is

$$u^2-2\ell u+br^2-2(\gamma-p_0v_y)r+v_y^2,\qquad v_y=y_0/p_2-p_0r_0.$$

After $r=5151/20000-3(u-151/200)+t$, its Hessian divided by two is

$$\begin{pmatrix}187/25&-54/25\\-54/25&18/25\end{pmatrix},\qquad\det=18/25>0.$$

Thus it is jointly convex in (u,t).

In the source-moment price the coefficients of the sixth cumulant and the projection allowance are respectively

$$A=\frac{22262234067601129}{2500000000000000000000}>0,\qquad B=\frac{172316040932398871}{2500000000000000000000}>0.$$

The fresh source bounds therefore give a price consisting of affine terms in each source coordinate, plus

$$\left[A\left\{\frac{194}{\sqrt{10}}-\frac{633}{9}(z-2/5)\right\}+\frac{9}{4}Bz^{3/2}\right]\kappa^{3/2}+B\kappa^2.$$

The coefficient in braces is positive for $2/5 \le z \le 1$. Consequently the price is separately convex in κ and z. Each affine parameter rectangle (u,t,κ,z) is therefore bounded by its 16 vertex prices. Independent exact-profile directed evaluation at 256 bits gives, respectively,

$$0.563356280368, \qquad 0.563648547036, \qquad 0.563686934863$$

as rational upper bounds for the three vertex maxima. These are strictly below the respective table caps. □

The restrictions on t are restrictions on the actual projection norm r. These moving bands do not cover every r at their listed amplitudes and do not by themselves bound the global operator norm.

Proposition 25 (Further certified source regions)**.** *For $b = 18/25$, use the physical parameters u, r, κ of an odd Gaussian sign pair. In every Gaussian dimension and for every admissible cubic tensor shape,*

$$u \in [341/500, 1381/2000], \quad 0 \le \kappa \le 1, \quad r \ge 1/5$$
$$\implies \quad \langle T_b f, g\rangle \le \frac{563371}{10^6} < \frac{40}{71}.$$

There are also the following full-even-block bounds. Put

$$t = r - \frac{389}{1000} + \frac{9}{5}\left(u - \frac{7}{10}\right).$$

Under the common restriction $-1/200 \le t \le 1/50$, the bounds are

u	κ	$\langle T_b f, g\rangle$ *at most*
$[.69, .70]$	$[0, 1]$	$551/1000$
$[.70, .72]$	$[0, 1]$	$68/125$
$[.70, .72]$	$[0, 2]$	$451/800$

The first two table caps are below $40/71$; the last is 0.56375. Every finite decimal here denotes its exact rational value.

Certificate proof. On $0 \le \kappa \le 1$, the accepted Stein row prices give

$$\mathbb{E}|Q| \le \frac{830643632499}{10^{12}} < \rho_1 := \frac{207661}{250000}.$$

The support–source comparison with this cap, $a = 9/5$, and $\lambda = 0$ has directed right side below $1.649920196798 < 1649921/10^6$. Thus every such physical source satisfies

$$r \le \frac{1649921}{10^6} - \frac{9}{5}u.$$

The two accepted rank-one R3 profiles cover the source intervals

$$[1/5, 199/500], \qquad [199/500, 1649921/10^6 - (9/5)u].$$

Their whole-line local proofs use constant reserve 1/50000 and sixth-degree reserve 1/5000000. Independent replay accepts all 21873 compact scalar leaves and the exact tails starting at64. Exact $y(r)$ transport and convex endpoint pricing on the two source polygons give respective maxima below 0.563249470254 and 0.563370758117, proving the first cap. The complete replay is recorded in `COMPLETE_RESTRICTED_DOMAIN_REPLAY.json`. The outer extension is recorded in `independent_finer_amplitude_extension.json`.

For the table, the accepted full-even profile uses constant reserve 1/1000 and sixth-degree reserve $1/10^6$. Its independent whole-line replay checks 56194 partition events, 27636 accepted leaves and 462 exact physical-ellipsoid exclusions, then glues the exact tail beyond64. The six matrix gates and $y(r)$ transport are exact. Let $a_r = b - E_{00} + E_{02}^2/E_{22} < 0$. On each (u, t) rectangle, product-corner interpolation incurs at most

$$(-a_r)\{(9/5)^2(\Delta u)^2 + (\Delta t)^2\}/4.$$

This follows because the mixed term is interpolated exactly and the variance of the affine source is $(9/5)^2 \operatorname{Var}(u) + \operatorname{Var}(t)$; the positive u^2 term only improves the comparison. Source prices are separately convex in κ and the shared shape. All48 vertex prices, including these reserves, have independent directed maxima below 0.550507802111, 0.543476058131, and 0.563687366986. They prove the table caps; the record is `general_r3_source_bands_audit.json`. □

The first conclusion retains both $\kappa \le 1$ and $r \ge 1/5$. The table retains its restrictions on t and κ. Their union is a collection of physical-source regions, and does not establish complete amplitude coverage or a global operator norm bound.

Proposition 26 (Exact transport to a smaller cubic coefficient)**.** *Freeze a valid physical Gaussian profile at $b_0 > 0$, including its whole-line local and scalar majorants. For every $0 < b \le b_0$ all its positive semidefinite gates remain valid. At the same physical parameters $(u, r, \kappa, \text{shape})$, the resulting pairing upper price changes exactly by*

$$(b - b_0)r^2 + c_\beta \left(\frac{1}{b} - \frac{1}{b_0}\right).$$

Here $c_\beta = \beta^2$ for orthonormal longitudinal Hermites and $c_\beta = 6\beta^2$ for raw longitudinal Hermites in the R3 criterion. No new local Gaussian integration is required. In the first case the local potential changes only by its constant $\beta^2(1/b - 1/b_0)$; in the second it is unchanged.

Proof. Each gate at the smaller coefficient is the old gate plus a positive semidefinite diagonal matrix. For example, the raw even gate gains $\operatorname{diag}(0, (b_0 - b)/2)$, and the raw odd gates gain $\operatorname{diag}(0, (b_0 - b)/6)$ or $\operatorname{diag}(b_0 - b, 0)$. The normalized gates behave likewise; the independent gates $E \succeq pp^T$, $E \succeq 0$, $R \succeq 0$ do not change. The rank-one raw gate also weakens.

The only remaining appearances of b in the completed criterion are the source term br^2 and the longitudinal square-completion constant. The norm of the raw third Hermite is squared norm six, accounting for the displayed factor. All source residual and scalar majorant terms are independent of b. In the normalized native criterion the added local constant cancels out of the centered Stein target, so its Stein solution and all derivative inequalities remain the same. Its added constant is simply charged in the outer expectation. This proves the exact price identity. □

Uniform cell pricing and a global coefficient. Suppose an old valid uniform cap U_0 covers a physical source cell $r \in [r_-, r_+]$ with $0 \le r_- \le r_+$. Since $b - b_0 \le 0$, the transported uniform cap

$$U_0 + (b - b_0)r_-^2 + c_\beta(1/b - 1/b_0)$$

is immediately valid. For tighter endpoint repricing, the native source curvature changes to $\xi(b) = \xi(b_0) + b - b_0$, and the full-even R3 curvature changes to $a_r(b) = a_r(b_0) + b - b_0$. Their respective chord reserves must be updated to

$$(-\xi(b))_+(r_+ - r_-)^2/4, \qquad (-a_r(b))_+(r_+ - r_-)^2/4.$$

Every such fixed-profile cell cap is convex in $b > 0$. A fixed selection of profiles covering all physical cells therefore gives a convex maximum of caps. A norm certificate requires one coefficient b valid for the entire cover. Minimizing separately for each sign pair, at $b = \min\{b_0, \sqrt{c_\beta}/r\}$ when $r, c_\beta > 0$, does not prove a bound for a single operator.

Theorem 27 (Complete amplitude branches at one coefficient)**.** *Set $b_* = 7171/10000$. For every pair of odd Gaussian signs f, g, in every finite Gaussian dimension, put $u = \|P_1((f+g)/2)\|_2$. Then*

$$\langle f, T_{b_*} g\rangle \le \begin{cases} 56338/10^5, & 0 \le u \le 1381/2000, \\ 563315250991/10^{12}, & 31/40 \le u \le \sqrt{2/\pi}. \end{cases} \qquad < \frac{40}{71}.$$

Both conclusions include every physical source amplitude, cumulant and cubic tensor shape.

Certificate proof. All the profiles here use the single coefficient b_*. The elementary energy estimate with $\rho = 847/1000$ gives

$$\langle f, T_{b_*} g\rangle \le u^2 + b_*\rho^2 \le \frac{7042043}{12500000} < \frac{40}{71} \quad (0 \le u \le 541/2500),$$

where the displayed cap conservatively uses $b = 18/25$. Four previously replayed complete native frames, transported by Proposition 26, cover $[.14, .42]$, $[.30, .485]$, $[.40, .54]$ and $[.50, .638]$. Their outward caps are respectively below .561498051269, .562840841523, .563045009361 and .563207127166. The intervals overlap each other and the elementary branch.

The exact native-frame union on $[.60, .682]$ consists of 30268 accepted cells and has maximum rational cap $563067382411/10^{12}$. On $[.682, .6905]$, 9661 accepted native cells and 41 cells covered by the transported R3 profile have maximum cap $563379835873/10^{12} < 56338/10^5$. The R3 replacement uses the independently replayed $11470 + 88$ compact leaves and exact tail beyond64, with constant and sixth-degree reserves $1/50000$ and $1/5000000$.

For completeness of these source unions, the support–source proof of (35) applies with the freshly proved $\rho = .847$. For each amplitude subinterval $[u_-, u_+]$, its source grid covers

$$0 \le r \le R_{.847}(u_-),$$

an outward bound on the entire physical source range because $R_{.847}$ is decreasing. The cumulant intervals partition $[0, 90]$. Native prices are affine in the variance shape $s \in [0, 1]$; the R3 coordinate is exactly $z = 2/5 + (3/5)s \in [2/5, 1]$. Every cell has either one complete native bound or one same-coefficient R3 bound. Exact positive source chords and quadratic source maxima, including interior vertices, cover its full continuous interior. Independent price replays check 2240 and 1344 native endpoint prices in these two unions, all selected profile hashes, and the sixteen R3 replacement corners. A separate exact verifier checks all $112 + 42$ amplitude–cumulant groups and every gap-free source partition. This proves the first branch.

The upper native union on $[.775, .797885]$ has 5096 cells and maximum cap $563315250991/10^{12}$; its independent replay checks 1568 endpoint prices and the same exhaustive physical-source partitions. Finally $u \le \sqrt{2/\pi} < .797885$. The complete union records are `native_lower_base_b7171_target_union.json`, `native_R3_b7171_u682_6905_union.json`, and `native_upper_b7171_target_union.json`; their frozen input hashes are checked in `same_b_native_R3_glue_audit.json` and the independent native price replay ledgers. □

The uncovered amplitude range for this theorem alone is $1381/2000 < u < 31/40$. The theorem is an upper bound for the indicated pairings, and does not by itself bound the unrestricted operator norm.

Lemma 28 (Exact pricing on physical source polygons)**.** *Fix an amplitude interval $[u_-, u_+]$, a source interval $[r_-, r_+]$, and a cumulant interval. Suppose that, throughout the cumulant interval and for every allowed shape, proved scalar inequalities imply*

$$a_j u + r \le C_j, \qquad a_j > 0 \quad (1 \le j \le m).$$

Let K be the intersection of these half-planes with the amplitude–source rectangle, after any separately proved coordinate restrictions. For rational inputs its vertices and the maximum on K of

$$F(u, r) = u^2 + L_u u + g r^2 + L_r r + C$$

are computable using rational arithmetic. For a nonempty K, it suffices to check its vertices and, on each edge, the stationary point in the edge interior when the restricted quadratic is concave. This includes segments and singletons. An empty intersection contains no physical source.

For a native certificate interpolated in r, first split K at every source knot. On each resulting piece apply this rule to each of the four cumulant–shape corner bounds, and take the maximum of the resulting bounds. A fixed nonnegative contact multiplier may be included by changing only L_u, L_r, C.

Proof. Every defining half-plane contains every physical source in the nominal rectangle, so intersection and coordinate clipping preserve all physical points. Rational line intersections give all vertices, retaining points on the boundaries. If a full-dimensional polygon had a maximizing point in its interior, sufficiently small horizontal displacements would stay in the polygon. Their two objective values have average $F(u, r) + \delta^2$, contradicting maximality. Thus its maximum lies on its boundary. On the edge $(u, r) = (u_0, r_0) + t(\Delta u, \Delta r)$, $0 \le t \le 1$, the quadratic coefficient is $(\Delta u)^2 + g(\Delta r)^2$. Endpoints suffice when this coefficient is nonnegative; otherwise also check the unique stationary point if it lies in $(0, 1)$. All these operations are rational. The same calculation covers the lower-dimensional cases.

A native source chord is valid only between successive saved knots. Splitting at every knot therefore retains its validity on each closed piece; shared boundaries have both valid endpoint bounds. For each fixed cumulant–shape corner the chord changes only the linear and constant coefficients in r. The already proved corner inequalities and positive interpolation uniformly bound the whole cumulant–shape cell. Taking the maximum over the four corner prices and all source pieces gives a uniform bound. The scalar row chosen at a source endpoint is fixed throughout its cumulant–shape cell; different endpoints may use different globally valid scalar rows.

For a contact with parameters a, B, put $d^2 = 6B^2$ and let A be its uniform scalar price. The nonnegative multiplier λ adds

$$-\lambda a u - \lambda r + \lambda \left(A + \frac{\beta d}{b} \right) + \frac{\lambda^2 d^2}{4b}.$$

Here β is the normalized native Hermite coefficient; for a raw Hermite R3 profile the product βd is $6\beta_{\mathrm{raw}} B$. The coefficient of u^2 remains one, so the same boundary computation applies. The multiplier and scalar contact are fixed on the whole cell. Selecting any one of several proved upper bounds on each cell preserves validity. □

Uniform source planes. For the accepted even sextic scalar rows, the moment price is affine in shape and convex in cumulant: the sextic coefficient is nonnegative and the coefficient multiplying $\kappa^{3/2}$ is positive throughout the shape interval. Hence the largest of the four endpoint prices is a uniform C_j. A single fixed scalar row is used for each plane over the whole cumulant interval. Taking the smallest uniform price among several such rows is valid; taking separate minima at the four corners would require a separate justification.

Proposition 29 (Base scalar certificate at .713)**.** *Set* $\widehat{b} = 713/1000$*. Then*

$$\mathcal{N}(\widehat{b}) \le \widehat{B} := \frac{35245449}{62500000} = .563927184.$$

Consequently the explicit reduced rational matrix sequence has actual limit

$$\lim_{n\to\infty} G(H_n(\widehat{b})) = \frac{1}{\mathcal{N}(\widehat{b})} \ge \frac{62500000}{35245449} = 1.7732785869744488\ldots.$$

The rational reciprocal is an unconditional lower bound for $G_{\mathbb{R}}$*. The scalar certificate does not assert* $\mathcal{N}(\widehat{b}) = \widehat{B}$*.*

Certificate proof. All branches use the same coefficient $\widehat{b}$. The independently verified lower-amplitude ledger covers $u \in [0, 1381/2000]$ with bound $563295214999/10^{12} < \widehat{B}$. It combines the low-energy branch, four transported regional frames and two complete native–R3 source unions. The exact transfer from $\bar{b} = 7171/10000$ is

$$(\widehat{b} - \bar{b})r^2 + \beta^2_{\mathrm{norm}}(1/\widehat{b} - 1/\bar{b}),$$

where $\beta^2_{\mathrm{norm}} = 6\beta^2_{\mathrm{raw}}$ for an R3 profile. The selected source unions retain the nonpositive first term; the spectral gates remain valid at the smaller coefficient. The lower ledger checks all 39929 native cells and 41 R3 cells in these two unions, along with the four earlier regional frames.

The original upper cover has 25 amplitude bands on $[1381/2000, 159577/200000]$ and 56 cumulant bands on $[0, 90]$. The endpoint $159577/200000$ exceeds the maximum physical amplitude $\sqrt{2/\pi}$. Every one of its 1400 groups covers every physical source from zero to an outward $R_{.847}(u_-)$ endpoint, with all shapes included. Native shape $s \in [0, 1]$ corresponds exactly to $z = 2/5 + (3/5)s$ for the R3 prices. Directed checks verify all 25 original source-grid upper endpoints and 250 inverse clips. Nineteen initial R3 profiles give 1064 cumulant-band constants, checked at all 4256 cumulant–shape corners. Seven native tables have 11648 independently checked baseline endpoint prices, followed by the exact transfer above. Their source quadratics also gain $\widehat{b} - \bar{b}$.

Each cumulant band has ten proved uniform positive-head source planes. The 560 planes and their moment corner prices are independently verified. Of the original 255360 source cells, 253282 retain upper prices and 2078 are exactly empty. The producer-free exact replay checks every original interval, selected polygon or containing rectangle, native radial knot and concave edge stationary point, using Lemma 28. It reconstructs all selected contact and ordinary prices and accounts for every original cell exactly once.

Three additional independently accepted whole-line profiles strengthen 1069 prices on the identical cover. An exact amplitude refinement then replaces 2814 difficult parent cells by 27340 child intervals: 20749 are nonempty and 6591 are exactly empty. The other 250468 price cells remain unchanged. Each child inherits its parent's proved source restrictions; its clipped polygon and selected price are checked independently. The old parent bound is retained as an option. Five further accepted whole-line profiles then strengthen prices on this same refined cover in three successive monotone stages, changing 2397, 27889 and 2428 prices respectively. Their 224, 448 and 448 new

cumulant–shape corner prices are independently recomputed. The resulting 27-profile cover has 271217 price cells and 8669 exact empty cells.

Finally, introduce a 68-point cumulant grid while retaining every previous bound. Only 364 difficult parents are split, giving an exact partition into 1794 cumulant children. Of these, 1726 retain prices and 68 are exactly empty; the remaining 270853 price cells are unchanged. Seven native endpoint tables on this finer grid have 13936 independently checked baseline corner prices, transported exactly to $\widehat{b}$. Fourteen new cumulant bands use ten fixed positive-head planes each, chosen from the already proved 60-row analytic bank. All 140 planes are independently checked at their 560 moment corners. Their intersections contain every physical child point. An exact replay verifies every child cumulant partition and complete clipped polygon. It reconstructs 1110 newly selected native prices, splitting every radial knot and including every concave edge stationary point; the other 616 nonempty children retain their valid parent bounds. No new R3 constant is needed in this last refinement.

Two final independently accepted whole-line profiles are added on this identical cumulant-refined cover. They strengthen 691 and 343 prices respectively, while retaining every prior bound. Each new profile has 280 independently recomputed cumulant–shape corner prices on the mixed collection of original and refined bands. An exact replay reconstructs every changed polygonal or rectangular price and verifies that all other cell records and all empty cells are unchanged.

The final complete upper cover thus contains 272579 price cells and 8737 exact empty cells. Its maximum recorded price is

$$\frac{4511417459}{8 \cdot 10^9} < \widehat{B}.$$

The frozen source is `global_upper_b713_native_kappa68_twentyninebank_envelope.json.gz`, with SHA-256 `ea763375a19e63a49174bcf18975c948ab188f993edfb471014cbb2bfec04d2d`. The final independent reconstruction is `fullupper_b713_native_kappa68_twentyninebank_monotone_audit.json`; it binds the complete chain of accepted parent covers and replay hashes.

These upper and lower branches therefore bound every odd sign pairing in every finite Gaussian dimension. Negating one sign gives the absolute norm bound. The established Gaussian-to-matrix convergence and exact ratio-preserving reductions give the actual matrix limit and lower bound. □

Theorem 30 (Complete transported scalar certificate)**.** *Set* $\widehat{b} = 711/1000$*. Then*

$$\mathcal{N}(\widehat{b}) \le \widehat{B} := \frac{112782853}{2 \cdot 10^8} = .563914265.$$

Consequently the explicit reduced rational matrix sequence has actual limit

$$\lim_{n\to\infty} G(H_n(\widehat{b})) = \frac{1}{\mathcal{N}(\widehat{b})} \ge \frac{2 \cdot 10^8}{112782853} = 1.7733192119195636\ldots.$$

The rational reciprocal is an unconditional lower bound for $G_{\mathbb{R}}$*; the norm need not equal its certified cap.*

Coefficient transport. Retain the complete upper physical partition and every selected frozen criterion from Proposition 29, at $b_0 = 713/1000$. Amplitude, source, cumulant, shape and all physical source restrictions are independent of the operator coefficient. All 8737 empty cells therefore remain empty.

For an ordinary frozen criterion let β_i be its normalized longitudinal coefficient: it is the recorded native coefficient, or $\sqrt{6}$ times the recorded raw R3 coefficient. Its pointwise price changes by

$$(\widehat{b} - b_0)r^2 + \beta_i^2(1/\widehat{b} - 1/b_0).$$

For a fixed head-contact criterion, with multiplier $\lambda_i \geq 0$ and head $a_c Z + B_c H_3(Z)$, put $d_c = \sqrt{6} B_c$. Completing the same longitudinal square gives instead

$$(\widehat{b} - b_0) r^2 + (\beta_i + \lambda_i d_c/2)^2 (1/\widehat{b} - 1/b_0).$$

Thus the ordinary case is included by taking $\lambda_i = 0$. These identities follow by grouping the longitudinal cost β_i^2/b, its contact cross term $\lambda_i \beta_i d_c / b$ and the contact square cost $\lambda_i^2 d_c^2/(4b)$. All other frozen prices are unchanged. All spectral matrix gates are checked at the smaller coefficient.

Let a_i be the accepted old upper price and $r_{i,-}$ its source cell's lower endpoint. Since $\widehat{b} - b_0 < 0$ and every physical point in that cell satisfies $r \geq r_{i,-} \geq 0$, a valid new uniform price is

$$a_i + (\widehat{b} - b_0) r_{i,-}^2 + (\beta_i + \lambda_i d_c/2)^2 (1/\widehat{b} - 1/b_0).$$

An inherited parent bound is treated in precisely the same way, using its original frozen criterion and multiplier; it already bounds every physical child point. The coefficient $\widehat{b}$ is fixed throughout all cells and is not chosen separately for different sign pairs.

An independent producer-free replay reconstructs all 43 exact profile bindings, verifies the target spectral gates and evaluates these shifts on every one of the 272579 price cells. The final assigned mosaic has 272557 ordinary cells and 22 R3 contact cells. Every one of their transport shifts is rational and is computed exactly. The maximum rounded shifted price is

$$\frac{563914263671}{10^{12}} < \widehat{B}.$$

The source is the hash-bound `assigned_twenty-nine_coefficient_scan.json`; the exhaustive independent cell replay is `assigned_b711_shifted_cells_audit.json`. The separately verified complete lower-amplitude branch at the same coefficient, $u \in [0, 1381/2000]$, has upper bound $563307766601/10^{12} < \widehat{B}$. It includes the elementary branch, four earlier regional frames and both complete lower source unions. Together the transported upper and lower covers contain every physical odd sign pair in every finite Gaussian dimension. Negating one sign gives the absolute norm bound. Gaussian-to-matrix convergence and the exact ratio-preserving reductions give the stated actual matrix limit. □

11 Consequences for search and reflection representations

The sequence lies in the full absolute doubly stochastic class. Formula (14) now performs that conversion. Evaluate the copies on their smaller seed, merge proportional rows and columns with their total absolute weights, and quotient signed row/column permutations.

At fixed order, decompose $|B|$ into a convex combination of permutation matrices and restore the fixed entry signs, giving signed permutations with simplex weights and no cancellation. A signed left permutation fixes the first term to the identity. Disconnected support ratios are C-weighted component averages, so keep a best component. A balanced cycle-sign support gauges to a nonnegative matrix and has ratio one. For two signed permutations with weights $t, 1-t$, each nontrivial component is an alternating cycle on $k \geq 2$ rows and k columns. A frustrated cycle has $C = k - 2\min(t, 1-t)$ and

$$V \leq k \sqrt{1 - 4t(1-t)\sin^2(\pi/(2k))}.$$

To see the latter bound, gauge its matrix to $tI + (1-t)S$, where the signed cyclic permutation has $S^k = -I$, and use $V \leq k \| \cdot \|_{\mathrm{op}}$. For $0 \leq t \leq 1/2$ the squared ratio is increasing, since

$k\sin^2(\pi/(2k)) \le 1$. Its maximum is at most $k\cos(\pi/(2k))/(k-1)$, which is $\sqrt{2}$ at $k=2$ and at most $3/2$ for $k \ge 3$. Thus a search near 1.775 needs frustrated cycles and at least three permutation terms; this does not assert that three suffice.

A weighted-reflection representative is also explicit. Since $|H_n|$ is doubly stochastic, $\|H_n\|_{\mathrm{op}} \le 1$. If its order is M_n, put

$$U_n = \begin{pmatrix} H_n & (I-H_n^2)^{1/2} \\ (I-H_n^2)^{1/2} & -H_n \end{pmatrix}, \quad P_n = (I-U_n)/2, \quad D_n = \mathrm{diag}((2/M_n)I_{M_n}, 0).$$

Then $P_n = P_n^T = P_n^2$, its diagonal is $1/2$, and

$$D_n - 2D_n^{1/2} P_n D_n^{1/2} = (2/M_n)H_n \oplus 0.$$

Its ratio is unchanged. Zero auxiliary weights are essential to this universal representation. Strictly equal positive weights on every coordinate would give an orthogonal matrix, whose ratio is at most $\pi/2$, and would wrongly exclude the target. Indeed, for an orthogonal matrix O of order m, $V(O) \le m$. Gaussian signs of Z_i and $(O^T Z)_j$ give

$$C(O) \ge \frac{2}{\pi}\sum_{ij} O_{ij}\arcsin(O_{ij}) \ge \frac{2}{\pi}\sum_{ij} O_{ij}^2 = \frac{2m}{\pi},$$

so $G(O) \le \pi/2$.

References

[1] R. Saha, A. Li, A. Xue, S. Chaudhuri, A. Klivans, P. K. Kothari and R. Meka, *New Lower and Upper Bounds for the Grothendieck Constant*, arXiv:2608.11158v2, 2026.

[2] Retained project report, *Real Grothendieck continuation*, September 30, 2026, Theorem 3 and its inherited analytic premises.

[3] Retained project report, *Matrix invariance and stochastic normalization*, September 30, 2026; independently reconstructed reductions in this continuation.

[4] Recovered matrix-mining sign certificates and *Baseline and transfer review*, October 1, 2026, included in the source package. The fixed-coefficient sign witnesses have fresh directed numerical replays; the historical global norm cover remains a separate premise.

[5] *Grothendieck 177128*, preceding delivered note and source package, October 1, 2026. Its endpoint depends on the cited Song–Yue uniform norm theorem and the stated nonlinear-cover scope.

[6] S. Heilman, C. Jones and G. Malavolta, *Sharper Bounds for the Complex Grothendieck Constant*, arXiv:2609.25462v1 (2026), equation (8.12). The tensor identities used here are also freshly checked by connected Wick diagrams.

A Exact tensor comparison

We give the finite acceptance rules for (26). Set $x = 1-y$ and $z = 1-a$. Each of 84 leaves specifies a closed rational rectangle in $[0,1]^2$, one of the upper bounds (24)–(25), and rational tangent parameters. For $t > 0$ the inequalities

$$\sqrt{X} \le (X+t^2)/(2t), \qquad \sqrt{XY} \le (tX + Y/t)/2$$

replace square roots by polynomial upper bounds. For a cube $X^{3/2}$ multiply the first inequality by $X \ge 0$. Choose rational $y_0 \in [0,1]$ and $l_0 > 0$ with $l_0^2 \ge 1 + 3y_0^2/2$. Cauchy–Schwarz gives

$$\sqrt{d} \ge L(y) = (1 + 3y_0 y/2)/l_0.$$

The bracket $[(K + 422/15)d - 211/3]$ is positive, since its minimum is $K - 211/5 > 0$. It is enough to show that the selected polynomial upper bound for $\mathcal{M}$ or $\mathcal{J}$ is at most this bracket times $L(y)$.

The dedicated equality-corner rectangle is $x, z \in [0, 2^{-19}]$. It uses $(p, q) = (3/4, -1)$ and $L(y) = (2+3y)/\sqrt{10}$. Here $(1-y^2)^{3/2} \le (1-y^2)/512$; tangent parameters $7/4$ and 1 handle the remaining roots. This retains equality at the corner.

Each difference polynomial has degree at most $(4,4)$ over $\mathbb{Q}(\sqrt{10})$. After substitution of the affine rectangle coordinates, let c_{kl} be its power coefficients. Its Bernstein coefficients are

$$b_{ij} = \sum_{k \le i,\, l \le j} c_{kl} \frac{\binom{i}{k}}{\binom{4}{k}} \frac{\binom{j}{l}}{\binom{4}{l}}, \qquad 0 \le i, j \le 4.$$

Every coefficient is nonnegative. The sign of $r + s\sqrt{10}$ is decided by rational sign cases and comparisons of r^2 with $10s^2$. Since the Bernstein basis is nonnegative and sums to one, the 25 checks prove the entire rectangle. The leaf counts are 21, 36 and 7 for the first three $\mathcal{J}$ bounds, 1 and 18 for the two Gram branches, and 1 for the dedicated corner: 84 rectangles and 2100 coefficients.

The rectangles lie within the square, have pairwise disjoint interiors, and have exact total area one. Their finite closed union covers the square: an omitted point would have a relatively open neighborhood of positive area. This checks every boundary as well as the interior. Two independent programs reconstruct the polynomials from the formulas. Neither imports the producer's coefficient lists or the older 99-rectangle data.

B Whole-line scalar certificates

All parameters of (31) and (30) are frozen in exact rational row files. The rows and complete cumulant cover are shown below; displayed prices are rounded upward, while acceptance uses directed rational enclosures.

Row	Cumulant interval	Corner price upper
0	$[0, 39/100]$	.825942224105
2	$[39/100, 39/25]$	.839598147439
3	$[39/25, 4]$	.847061071421
4	$[4, 34/5]$	.847358899934
$k8$	$[34/5, 48/5]$	.848520760642
5	$[48/5, 76/5]$	.848507934475
6	$[76/5, 28]$	.846999761374
7	$[28, 90]$	.838404593017

For each row, 512-bit outward Arb arithmetic and fourth-order Taylor bounds verify the entire compact interval $[0, 64]$ on an adaptive rational partition. Mills' ratio is enclosed at the endpoints, and $R' = xR - 1$ generates the needed Taylor coefficients. For $x \ge 64$, $0 \le 1 - xR(x) \le x^{-2}$ and convexity of the square reduce the comparison to two rational polynomial tests. Clearing the positive denominators and substituting $x = 64 + t$ gives nonnegative coefficients and positive constants, proving the complete unbounded tail. Evenness supplies the negative half-line.

The independent verifier uses direct Chebyshev recurrence, 768-bit arithmetic and sixth-order Taylor bounds on $[0, 80]$, followed by a separately reconstructed exact tail at 80. It also independently assembles all 32 prices in (32). A further checker reconstructs the 16 tail comparisons and

32 corner prices without importing either scalar implementation. Complete comparisons, rather than sampled maxima, are the acceptance criterion.

C Polynomial certificate for the cubic source gate

The third numerator in Proposition 10 is explicitly

$$\begin{aligned} P_3 = {} & A^6/8 + A^5B(3w/2 + 21/4) \\ & + A^4B^2(57w^2/8 + 195w/4 + 84) \\ & + A^3B^3(17w^3 + 339w^2/2 + 540w + 504) \\ & + A^2B^4(171w^4/8 + 549w^3/2 + 1194w^2 + 1728w) \\ & + AB^5(27w^5/2 + 837w^4/4 + 1116w^3 + 1980w^2) \\ & + B^6(27w^6/8 + 243w^5/4 + 378w^4 + 756w^3). \end{aligned}$$

Every coefficient is nonnegative for $A > 0$, $B, w \geq 0$. Thus this finite identity establishes the derivative sign on the entire half-line; no numerical Gaussian quadrature is used. As an exact example, take $A = 97/40$, $B = 1/6$. Then

$$a = 117/40, \quad d = \sqrt{6}/6, \quad (q_0, q_2, q_4, q_6) = \left(\frac{331}{120}, \frac{20}{97}, -\frac{1096000}{265587843}, \frac{106274560000}{727184560303017}\right).$$

At $\kappa = 53/100$, $u = 151/200$, $v = -1/4$, the two variance endpoints give the directed source caps

$$r < 252844293/10^9 \quad (V = 7\kappa/45), \qquad r < 250945617/10^9 \quad (V = 2\kappa/9).$$

The same affine shape interpolation applies between them. These conditional examples illustrate the gate; $v = -1/4$ is not assumed for arbitrary optimizing signs. Independent exact polynomial and directed price implementations verify the recurrence, integration constants and two rational caps.

D Complete continuous regional certificates

For $T = P_1 - (18/25)P_3$, odd signs f, g, and $u = \|P_1((f+g)/2)\|_2$, the following complete certificates give the displayed bounds on $\langle f, Tg\rangle$:

amplitude interval	local boxes	scalar rows	rational upper cap
$[7/50, 21/50]$	63681	24	$561523/10^6$
$[3/10, 97/200]$	61829	24	$562857/10^6$
$[2/5, 27/50]$	58841	24	$563039/10^6$
$[1/2, 319/500]$	151929	24	$563266/10^6$
$[3/5, 331/500]$	153593	24	$563302/10^6$
$[331/500, 341/500]$	627497	24	$563367/10^6$
$[77/100, 159577/200000]$	53541	24	$563334/10^6$

Every cap is strictly below $40/71$. The respective directed totals are below 0.561522410, 0.562856131, 0.563038808, 0.563265432, 0.563301723, 0.563366577, and 0.563333793. The first profile uses the universal transverse bound $r \leq 849/1000$ and therefore remains valid throughout its extended amplitude interval. The other extensions are checked against the outward source

endpoint. The outer polynomial $u^2 - 2\ell u + C$ is convex, so rational endpoint checks cover each entire interval.

Each certificate has an optimizer-free, directed-arithmetic replay of all physical moment boxes, scalar compact leaves, infinite scalar tails, the five matrix gates, and the continuous cumulant, shape, and source interpolations. Infinite tails explicitly begin beyond every target and regression hinge. Local height and scalar coefficient repairs are fully priced. These are regional norm bounds; amplitudes outside the stated coverage still require certification before a new unrestricted norm cap or Grothendieck lower endpoint follows. The primary union is $[.14, .682] \cup [.77, \sqrt{2/\pi}]$. The separately proved low-energy branch covers $[0, .14]$. Thus the presently unclosed target amplitude interval is $(.682, .77)$. The earlier, tighter bound $9/16$ on $[.79, \sqrt{2/\pi}]$ and its original portable certificate are retained as supplementary results.

E Original-degree accounting for the sign witness

In this appendix $H_j = \mathrm{He}_j/\sqrt{j!}$ denotes the orthonormal longitudinal Hermite polynomial. The central limit theorem for Y_N alone does not justify assigning it ordinary Hermite degree one. Each summand has original degree three. Here we verify all projections used in (18).

First replace the sign by a bounded smooth approximation with bounded derivatives. Expand in one transverse summand $H_3(X_i)/\sqrt{N}$. Orthogonality and parity give

$$\mathbb{E}[H_1 H_3] = \mathbb{E}[H_1 H_3^2] = \mathbb{E}[H_2 H_3] = 0.$$

Taylor expansion therefore makes the $H_1(X_i)$ coefficient $O(N^{-3/2})$ and the $H_2(X_i)$ coefficient $O(N^{-1})$. The $H_3(X_i)$ coefficient is $N^{-1/2}\mathbb{E}\partial_y f + O(N^{-3/2})$. Expand separately in distinct transverse coordinates for mixed terms. The complete list of original-degree-one and degree-three basis types is

Basis type	Coefficient order	Summed squared energy
$H_1(Z)$	$\to m_1$	retained
$H_1(X_i)$	$O(N^{-3/2})$	$O(N^{-2})$
$H_3(Z)$	$\to m_3$	retained
$H_2(Z)H_1(X_i)$	$O(N^{-3/2})$	$O(N^{-2})$
$H_1(Z)H_2(X_i)$	$O(N^{-1})$	$O(N^{-1})$
$H_1(Z)H_1(X_i)H_1(X_j)$	$O(N^{-3})$	$O(N^{-4})$
$H_3(X_i)$	$N^{-1/2}m_Y + o(N^{-1/2})$	$\to m_Y^2$
$H_2(X_i)H_1(X_j)$	$O(N^{-5/2})$	$O(N^{-3})$
$H_1(X_i)H_1(X_j)H_1(X_k)$	$O(N^{-9/2})$	$O(N^{-6})$

The indices i, j, k are distinct. For a fixed smooth approximation the orders are uniform in these indices; the last column counts every term of the displayed type. The ordinary central limit theorem and normal integration by parts identify the three retained moments. Reflection of Y_N preserves m_1, m_3 and changes m_Y to $-m_Y$. Thus the degree-one and degree-three cross-pairings tend respectively to m_1^2 and $m_3^2 - m_Y^2$.

Finally $q'(z) \geq 511/200 > 0$. Conditional on Y_N, the normal density and this derivative bound give a small-band estimate

$$\Pr(|Y_N + q(Z)| \leq \varepsilon) \leq \frac{2\varepsilon}{(511/200)\sqrt{2\pi}},$$

uniformly in N. The same holds for the reflected witness. Consequently the smooth approximation can be removed uniformly in L_2; orthogonal projection is an L_2 contraction. This completes

the limiting degree calculation. The exact witness parameters, directed compact integration and infinite-tail bounds are supplied in `reduced_search/operator/cubic720/README.md` and the adjacent `cubic720_norm_witness.json`.

F Reproduction, retained premises and proof status

The source archive contains the exact row inputs, tensor rectangles, complete-halfline checkers, formal Wick enumeration, implicit matrix construction, independent audits, and directed MPFR numerator and sign witness calculations. The main checks can be run from the archive root:

```
python reduced_search/energy/633/certify_wick_identity.py
python reduced_search/energy/633/discover_cover.py --replay
python reduced_search/energy/633/independent_replay.py
python reduced_search/energy/certify_stein_cubic_L1.py
python reduced_search/energy/audit_stein_L1_independent.py
python tonight/moments/certify_cubic_head_polynomial.py
python tonight/theory/certify_numerator_rate.py
python tonight/theory/certify_retained_finite_ratio.py
python tonight/theory/certify_saha_benchmark.py
python tonight/theory/certify_published_upper_rounding.py
python tonight/operator/cubic_L1_refinement/refit847/verify_cover.py
python tonight/audit/refit847/replay_complete_refit_cover.py
python tonight/regional_certificates/u014_042/code/verify_region.py
python tonight/regional_certificates/u030_0485/code/verify_region.py
python tonight/regional_certificates/u040_054/code/verify_region.py
python tonight/regional_certificates/u050_0638/code/verify_region.py
python tonight/regional_certificates/u060_0662/code/verify_region.py
python tonight/regional_certificates/u0662_0682/code/verify_region.py
python tonight/regional_certificates/u077_c/code/verify_region.py
python tonight/operator/projection/replay_R3_cell/verify_all.py
python tonight/operator/projection/replay_accepted_R3/verify_all.py --quick
python tonight/operator/preferred_global/verify_bundle.py
python reduced_search/operator/cubic7171/certify_cubic7171_norm_witness.py
python tonight/audit/replay_cubic7171_witness.py
python tonight/audit/replay_selected_cubic_witnesses.py
```

Python, python-flint and the system MPFR library suffice for these accepted checks; discovery programs have additional numerical dependencies. Files are bound by a SHA-256 manifest. Floating optimization identifies candidate inputs and is kept separate from proof outputs.

For $b = .72$, directed evaluations of (13) give

n	Certified vector numerator lower bound
32	.999159371292
128	.999947506710
1024	.999999179839

These bounds are unconditional numerator estimates. A denominator bound valid for all sign pairs in every dimension is still required before (16) gives the target lower endpoint.

The global cap in Theorem 30, inserted into (16) at $b = 711/1000$, gives

n	Unconditional lower bound for $G(H_n(.711))$
32	1.771831610393
128	1.773226593281
1024	1.773317765118

These displayed lower bounds converge to 200000000/112782853. The actual ratio converges to $1/\mathcal{N}(.711)$, which may be larger. All finite rounding losses, including the total entry error n^{-4}, are charged by the directed checker `evaluate_finite_ratio.py`.

For comparison, the retained .699 denominator premise already gives explicit finite members of the reduced sequence. Directed evaluation of (16), with that premise stated separately, yields

n	Conditional lower bound for $G(H_n(.699))$
32	1.771523050434
128	1.772908047308
1024	1.772998563604

These conditional finite lower bounds converge quantitatively to 1.773; the actual ratio limit remains $1/\mathcal{N}(.699)$. These numbers use the earlier scalar norm premise and do not reprove it. The very large matrices are specified implicitly by their analytic seed and exact cloning maps.

The inherited project estimate (1) remains an explicit premise. Its 55-interval cover and inherited integrations are not reconstructed by the present archive. The exact reductions, asymptotic sequence, coupled cumulant theorem, cubic absolute-moment theorem, and support–source propositions above have their own fresh proofs and do not depend on that missing reconstruction.

The conditional project lower endpoint is 1.773, at distance .002 from 1.775. The later 1.7712846852... family was weaker and does not replace it. The fixed .699 construction has a directed ceiling 1.77440101857...; the fresh .72 ceiling is 1.77543615103..., and the fresh .7171 ceiling is 1.775462496412. These ceilings bound the reciprocal-norm methods, not the unrestricted Grothendieck constant. The regional amplitude caps and the complete same-coefficient global envelope prove the fresh unconditional lower endpoint 1.7733192119.... They do not prove the stronger target norm estimate $\mathcal{N}(b) \leq 40/71$ at the remaining candidate coefficients.